Stochastic Contextual Dueling Bandits under Safety Constraints

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Abstract

Learning from pairwise preference feedback is common in interactive systems, where explicit numeric rewards are often unavailable. This becomes particularly challenging in safety-critical applications such as content recommendation for minors, where the learner must satisfy strict but unknown safety constraints at every interaction. However, existing safe online learning methods rely primarily on numerical rewards to estimate safety boundaries, rendering them inapplicable to preference-based feedback. To bridge this gap, we formalize the problem as stochastic contextual dueling bandits under safety constraints. We propose Safe-COLSTIM, a novel algorithm that integrates conservative safety mechanisms with optimism-driven learning. By coupling conservative safety enforcement with preference learning, Safe-COLSTIM enables efficient exploration from pairwise feedback while maintaining safety throughout learning. We prove that Safe-COLSTIM achieves sublinear regret and guarantees safety under standard regularity conditions, both when a strict safety margin exists and when the margin is unknown. Empirically, Safe-COLSTIM exhibits near-zero safety violations across a range of safety gap settings while maintaining competitive regret against state-of-the-art unconstrained baselines, demonstrating that strong safety guarantees can be achieved without sacrificing learning efficiency.

1 Introduction

Modern recommendation systems face a fundamental challenge: they must learn user preferences through interactive feedback while satisfying strict operational constraints imposed by regulations, platform policies, or ethical considerations. Consider short-video platforms like TikTok or YouTube Shorts. These systems often infer user preferences from implicit *comparisons*, for example, a user keeps watching one video rather than skipping to another. At the same time, they must enforce content-safety policies. When serving content to minors, *every* recommended video must meet predefined safety thresholds regardless of predicted engagement, since even a single violation can trigger regulatory penalties or cause direct harm. Another example is e-commerce platforms. These platforms provide personalized recommendations and learn from users' comparative choices (e.g., clicking one item over another). At the same time, they must ensure that bias across demographic groups remains within acceptable bounds at *every* interaction to comply with fairness requirements.

These applications share two key characteristics. First, they rely on *preference-based feedback* rather than explicit numerical ratings: user behavior reveals relative preferences through *comparative* signals, such as which of two items a user chooses when shown side by side. Second, they operate under *per-round safety constraints* that must be satisfied at every decision point, unlike cumulative budget constraints that allow occasional violations. Moreover, these safety constraints often have structures in terms of item attributes and *unknown* system parameters, for example, linear relationships between latent safety coefficients and item features. As a result, enforcing such strict per-round

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safety constraints fundamentally changes the nature of preference learning: the learner must infer user preferences from noisy pairwise comparisons while simultaneously guaranteeing safety at every decision point, despite not knowing the true safety boundary in advance. This dual requirement creates fundamental algorithmic and analytical challenges that are absent from standard preference-learning formulations.

Similar preference-learning problems are often modeled as *sequential decision-making problems*, which have been studied through several complementary online-learning frameworks. The multi-armed bandit (MAB) model [5] captures the exploration-exploitation trade-off when rewards are initially unknown, while the linear stochastic bandits [13] leverage feature representations by assuming outcomes depend linearly on unknown parameters, incorporating contextual information [12, 23]. When feedback is inherently comparative rather than numeric, the dueling bandit framework [28, 7] observes noisy pairwise outcomes instead of numerical rewards. More recently, contextual extensions such as Contextual Linear Stochastic Transitivity (CoLST) models [8] posit that each item's latent utility is linear in contextual features, and that observed comparisons arise from these utilities under stochastic noise.

However, enforcing strict safety guarantees in preference-based learning introduces nontrivial challenges that are not addressed by existing frameworks. Most prior work on safe bandits focuses on cumulative budget constraints [6, 3] or enforces conservative behavior relative to known safe baselines [26, 16], neither of which can guarantee safety at *every* decision point with *unknown* constraint parameters. While per-round safety constraints have been studied in linear bandits [4], these methods rely critically on the ability to directly observe numerical rewards, which are unavailable under preference feedback.

Consequently, estimating safety boundaries becomes inherently more difficult, as the learner must control *absolute* feasibility while observing only *relative* comparisons. The challenge is further compounded when safety constraints depend on the same unknown parameter governing preferences, requiring simultaneous learning of both user preferences and safety boundaries from the same comparative signal.

Our Contributions. To bridge this gap, we develop the first framework for contextual dueling bandits under per-round safety constraints, tackling the fundamental challenge of guaranteeing safety at every decision point using only relative preference feedback. Our main contributions are summarized as follows:

- **Stochastic contextual dueling bandits under linear safety constraints:** We formulate the problem of contextual dueling bandits under CoLST models subject to *per-round* linear safety constraints on *unknown* preference parameters. This setting explicitly captures safety-critical applications where certain actions must be avoided to ensure user safety or regulatory compliance.

- **A provably safe and efficient algorithm:** We propose Safe-COLSTIM, a safety-constrained algorithm that combines conservative safety mechanisms with optimism-driven preference learning. The algorithm employs a two-phase design, starting with a conservative pure-exploration phase and transitioning to an adaptive safe exploration-exploitation phase. We show that Safe-COLSTIM enforces per-round safety constraints throughout learning while enabling efficient preference learning from noisy pairwise feedback.

- **Regret guarantees under different safety regimes:** We establish sublinear regret bounds for Safe-COLSTIM. With a positive safety gap, Safe-COLSTIM achieves regret on the order of $\widetilde{\mathcal{O}}(d\sqrt{T})$, where d is the context dimension and T is the time horizon, matching the optimal regret of unconstrained baselines. In the harder case where the safety gap is zero or unknown, the regret scales as $\widetilde{\mathcal{O}}((dT)^{2/3})$. Under mild spectral conditions, we further obtain improved bounds that highlight fundamental trade-offs between safety constraints and learning efficiency.

- **Empirical validation:** Through extensive experiments on synthetic environments, we show that Safe-COLSTIM maintains zero safety violations across various safety gap settings while achieving regret performance comparable to unsafe baselines. These results confirm that strong safety guarantees can be achieved in practice without sacrificing learning efficiency.

2 Problem Setting

Notation. We use $[n]$ to represent the set $\{1, 2, \ldots, n\}$ for any positive integer n, and $1_{[\cdot]}$ denotes the indicator function. For a vector $x \in \mathbb{R}^d$, the Euclidean norm is written as $\|x\|_2$. When referring to a positive semi-definite matrix $A \in \mathbb{R}^{d\times d}$, we define the weighted 2-norm of a vector x with respect to A as $\|x\|_A = \sqrt{x^\top Ax}$. The minimum and maximum eigenvalues of a matrix A are denoted $\lambda_{\min}(A)$ and $\lambda_{\max}(A)$, respectively. For symmetric matrices A and B, we write $A \preceq B$ when $B - A$ is positive semi-definite. Lastly, we use the standard $\widetilde{\mathcal{O}}$ notation for asymptotic analysis, which ignores logarithmic factors.

We consider a contextual dueling bandit setting where a learner interacts with a user over T rounds. The learner chooses from a finite set of actions (or arms) indexed by $\mathcal{A} = \{1, 2, \ldots, n\}$. Each arm $i \in \mathcal{A}$ is characterized by a d-dimensional feature vector, encoding attributes such as video topic categories, content ratings, or product specifications. We assume the existence of a known global context space $X_0 \subset \mathbb{R}^d$, which contains all possible feature vectors associated with the arms. Throughout this work, we consider bounded contexts with $X_0 := \{x \in \mathbb{R}^d : \|x\|_2 \leq 1\}$. At each round $t \in \{1, \ldots, T\}$, the learner observes a contextual decision set $X_t = \{x_{t,1}, \ldots, x_{t,n}\} \subseteq X_0$, where each $x_{t,i}$ denotes the feature vector associated with arm i at round t. We further assume that the decision sets $\{X_t\}_{t\geq 1}$ are drawn i.i.d. from an unknown distribution $\mathcal{D}$ over X_0, capturing the stochastic nature of contextual features in real-world applications such as recommendation systems.

Preference Feedback Model. Instead of numerical rewards, the learner receives relative feedback. At round t, after observing the context set X_t, the learner selects a pair of arms $S_t = (i_t, j_t)$ and receives binary preference feedback $Y_t = 1_{[\,i_t \succ j_t\,]}$, where the preference relation $i_t \succ j_t$ means that arm i_t is preferred over arm j_t under the observed context X_t. Accordingly, $Y_t = 1$ if i_t is preferred to j_t, and $Y_t = 0$ otherwise.

We model pairwise preferences using a random utility framework underlying the Bradley–Terry–Luce (BTL) model [15]. Specifically, each arm i is associated with a latent (deterministic) utility $u_i \in \mathbb{R}$. Preference uncertainty is captured by additive random perturbations: the observed utility of arm i is $v_i = u_i + \epsilon_i$, where ϵ_i are i.i.d. random variables drawn from a perturbation distribution G^*. The learner prefers arm i over arm j if and only if $v_i \geq v_j$.

This construction implies that pairwise preferences depend only on the difference between latent utilities:

$$\mathbb{P}(i \succ j) = \mathbb{P}(v_i \geq v_j) = \mathbb{P}(u_i - u_j \geq \epsilon_j - \epsilon_i) =: F^*(u_i - u_j),$$

where F^* is the comparison function induced by the perturbation distribution G^*.

To incorporate contextual information, we adopt the Contextual Linear Stochastic Transitivity (CoLST) model [8], in which the latent utility of arm i at round t is a linear function of its feature vector:

$$u_i^*(X_t) = x_{t,i}^\top \theta^* = \langle x_{t,i}, \theta^* \rangle, \tag{1}$$

where $\theta^* \in \mathbb{R}^d$ is an unknown parameter vector.

Since preferences depend only on utility differences, we define the difference feature vector $z_{t,i,j} := x_{t,i} - x_{t,j}$, which implies $u_i^*(X_t) - u_j^*(X_t) = \langle z_{t,i,j}, \theta^* \rangle$. Consequently, the probability of preferring i_t over j_t for the selected arm pair (i_t, j_t) is

$$\mathbb{P}(Y_t = 1 \mid X_t) = F^*(\langle z_{t,i_t,j_t}, \theta^* \rangle)\,. \tag{2}$$

In this work, we focus on the logistic specification of F^*, i.e., $F^*(v) = 1/(1 + e^{-v})$, which arises when the perturbation distribution G^* is the standard Gumbel distribution. In this case, the difference of two i.i.d. Gumbel random variables follows a logistic distribution, recovering the classical BTL comparison function. More generally, the CoLST framework allows for other choices of G^*, each inducing a different comparison function F^*. Our theoretical analysis applies to any F^* satisfying mild regularity conditions (see Assumption 2.3).

Safety Constraint. The learning environment is subject to operational restrictions that limit the set of admissible actions. At each round, the learner is required to select an action from a *safe set*,

which partitions the action space into safe and unsafe regions according to an underlying feasibility constraint.

We model the safety requirement using the linear constraint

$$\theta^{*\top} Bx \leq c, \tag{3}$$

where $B \in \mathbb{R}^{d\times d}$ and $c > 0$ are known constants, and $\theta^* \in \mathbb{R}^d$ is an unknown parameter governing both preferences and safety. The corresponding (oracle) safe set is defined as

$$X_0^S := \big\{x \in X_0 : \theta^{*\top} Bx \leq c\big\}. \tag{4}$$

Although the learner observes contextual features and preference feedback, the quantity $\theta^{*\top} Bx$ is never directly revealed. As a result, the true safe set X_0^S cannot be identified exactly, and naïvely optimizing without accounting for parameter uncertainty may lead to unsafe decisions. To address this challenge, we adopt a *conservative safe learning* principle, selecting actions only if they can be certified to satisfy the safety constraint for all statistically plausible parameters. The construction of such conservative safe sets is deferred to later sections.

Regret. As described in the Preference Feedback Model, under the Contextual Linear Stochastic Transitivity (CoLST) framework, each arm i at round t admits a latent utility (see (1)). It is therefore natural to define the optimal arm at time t as the one with the highest latent utility among the arms that are feasible at that round. More specifically, we define the best arm at the current round t as

$$i^*(t) = \arg\max_{i\in\mathcal{A}_t} u_i^*(X_t),$$

where $\mathcal{A}_t$ denotes the set of arms that are deemed *safe* at time t. The set $\mathcal{A}_t$ is induced by a time-dependent conservative safe region constructed using uncertainty quantification, and is formally defined in Section 3 (see (8)).

Following standard notions in dueling bandits, we measure performance using the *average regret* incurred by selecting a pair of arms $S_t = (i_t, j_t)$ at round t. The instantaneous average regret is defined as

$$r_t^a(S_t, X_t) = \frac{u_{i^*(t)}^*(X_t) - u_{i_t}^*(X_t) + u_{i^*(t)}^*(X_t) - u_{j_t}^*(X_t)}{2}, \tag{5}$$

which captures the average utility loss of the selected pair relative to the optimal arm. Aggregating over the horizon T, the cumulative average regret is defined as

$$R_T^a\big((S_t)_{t\in[T]}\big) = \sum_{t=1}^{T} r_t^a(S_t, X_t).$$

Objective. The objective of the learner is to keep the cumulative average regret R_T^a as small as possible over a horizon of T rounds. At a minimum, we require that the learning algorithm achieves vanishing average regret, i.e., $R_T^a/T \to 0$ as $T \to \infty$.

In contrast to existing contextual dueling bandit formulations, we additionally impose a safety requirement on the learner. Specifically, the actions selected by the algorithm must be safe with high probability, meaning that for all $t \in [T]$, the selected actions satisfy $x_{t,i} \in X_0^S$. Throughout this paper, we use the term *regret* to refer to the cumulative average regret R_T^a defined in the previous section, and study algorithms that jointly minimize regret while respecting the safety constraint.

Model Assumptions. For our theoretical analysis, we adopt the following assumptions on the context space, the safety constraints, and the preference feedback model. These assumptions are all standard and widely used in the literature on contextual bandits and preference-based learning (e.g., [17, 24]).

Assumption 2.1 (Boundedness and Stochastic Informative Contexts)**.** The unknown parameter $\theta^* \in \mathbb{R}^d$ satisfies $\|\theta^*\|_2 \leq 1$, and the inner products are uniformly bounded as $\theta^{*\top} x \in [-1, 1]$ for all $x \in X_0$. Moreover, there exists a constant $\rho > 0$ such that the population covariance over a designated *exploration sampling rule* $\pi_{\exp}$ (specified in Algorithm 1, line 4) satisfies

$$\lambda_{\min}\big(\mathbb{E}_{X_t\sim\mathcal{D}}\,\mathbb{E}_{x\sim\pi_{\exp}(X_t)}[\,xx^\top\,]\big) \geq \rho > 0.$$

Assumption 2.2 (Non-empty Safe Set)**.** The decision set X_0 is a convex body in $\mathbb{R}^d$ containing the origin in its interior. Together with the assumption $c > 0$ in the safety constraint, this ensures that the safe decision set X_0^s is non-empty and contains the origin..

Assumption 2.3 (Regularity of the Comparison Function)**.** The comparison function F^* admits an L-Lipschitz continuous derivative for some $L > 0$:

$$\left|(F^*)'(x) - (F^*)'(y)\right| \leq L|x - y|, \quad \forall x, y \in \mathbb{R}.$$

Moreover, the derivative is uniformly bounded below in a neighborhood of the true parameter:

$$\mu := \inf\left\{(F^*)'(x^\top\theta) \mid \|x\| \leq 2,\ \|\theta - \theta^*\| \leq 1\right\} > 0.$$

This ensures F^* remains strictly monotone over relevant utility differences and prevents degeneracy in preference signals.

Assumption 2.4 (Known Safe Seed Set)**.** There exists a known finite seed set $\mathcal{S}_0 \subseteq X_0$ with $|\mathcal{S}_0| \geq \max(2, d)$ satisfying:

(i) (*Safety*) ${\theta^*}^\top Bx \leq c$ for all $x \in \mathcal{S}_0$.

(ii) (*Spanning*) $\lambda_{\min}\big(\frac{1}{|\mathcal{S}_0|}\sum_{x\in\mathcal{S}_0} xx^\top\big) \geq \rho_0 > 0$.

(iii) (*Persistence*) $\mathcal{S}_0 \subseteq X_t$ for every round $t \geq 1$ almost surely under the context distribution $\mathcal{D}$.

Remark. Condition (iii) reflects the practical reality that recommendation platforms typically include a small set of editorially curated safe items in every round's candidate pool (e.g., default safe content, vetted advertisements). The seed set $\mathcal{S}_0$ formalizes this operational structure and ensures that the algorithm always has access to a known-safe spanning set, regardless of the geometry of the remaining arms in X_t.

3 Safe-COLSTIM Algorithm

In this section, we introduce our Safe-COLSTIM algorithm. Unlike existing studies that focus on optimizing preference-based decisions, Safe-COLSTIM explicitly balances exploration, exploitation, and safety, which is crucial for safety-critical applications. The algorithm minimizes cumulative regret over time while ensuring that safety boundaries are respected. This is achieved through the principle of *optimism in the face of uncertainty (OFU)*, where the learner selects actions that are expected to perform well under the current estimate while maintaining safety. The detailed procedure is outlined in Algorithm 1, which covers both the exploration and exploitation phases, including how safety constraints are integrated into decision-making.

3.1 Algorithm Overview

In this section, we give a detailed description of the Safe-COLSTIM algorithm, whose pseudocode is summarized in Algorithm 1. The algorithm follows a two-phase structure, consisting of an initial *pure exploration phase* and a subsequent *safe exploration-exploitation phase*. The former is designed to initialize reliable statistical estimates, while the latter jointly balances preference learning, optimism-driven exploration, and safety enforcement.

Pure Exploration Phase. During the pure exploration phase (lines 1– 5 of Algorithm 1, rounds $t = 1, \ldots, \tau$), the learner samples arm pairs from a per-round *worst-case safe candidate set*

$$X^w \ := \ \left\{x \in X_t : \max_{\theta\in\Theta} \theta^\top Bx \leq c\right\} \ \cup \ \mathcal{S}_0, \tag{6}$$

where $\Theta := \{\theta \in \mathbb{R}^d : \|\theta\|_2 \leq S\}$ with $S \geq \|\theta^*\|_2$ a known upper bound (here $S = 1$ by Assumption 2.1). By Cauchy–Schwarz, $\max_{\theta\in\Theta} \theta^\top Bx = S\|Bx\|_2$, so any x satisfying $\|Bx\|_2 \leq c/S$ is safe under *every* $\theta \in \Theta$ (and in particular under θ^*). Combined with Assumption 2.4(i), this implies $X^w \subseteq X_0^S$ for all t. By Assumption 2.4(iii), $\mathcal{S}_0 \subseteq X^w$, so $|X^w| \geq |\mathcal{S}_0| \geq 2$ for every t. The exploration sampling rule π_{exp} denotes the marginal distribution over individual sampled arms induced by uniformly sampling two distinct arms without replacement from X^w.

The learner then observes the corresponding preference feedback $Y_t = 1_{[i_t \succ j_t]}$. The resulting comparison vectors $\{z_{s,i_s,j_s}\}_{s=1}^{\tau}$ collected during the pure exploration phase are used to form the initial Gram matrix $M_{\tau+1}$ (line 5).

Algorithm 1: Safe-COLSTIM

Input: Exploration length $\tau > 0$, perturbation distribution G with induced comparison (or cumulative distribution) function F, threshold $C_{\text{thresh}} > 0$, coupling probabilities $(p_t)_t$, confidence width constant $c_1 > 0$

1 **Pure exploration phase:**
2 **for** $t = 1, 2, \ldots, \tau$ **do**
3 Observe context vectors $X_t = (x_{t,1}, \ldots, x_{t,n})$ then form $X^w := \{x \in X_t : \max_{\theta \in \Theta} \theta^\top B x \leq c\} \cup \mathcal{S}_0$;
4 Sample (x_{t,i_t}, x_{t,j_t}) uniformly without replacement from X^w;
5 Get the corresponding preference feedback $Y_t = \mathbf{1}_{[i_t \succ j_t]}$ and set $z_{t,i_t,j_t} \leftarrow x_{t,i_t} - x_{t,j_t}$;
6 **end**
7 Set $M_{\tau+1} = \sum_{s=1}^{\tau} z_{s,i_s,j_s} z_{s,i_s,j_s}^T$;
8 **Safe exploration-exploitation phase:**
9 **for** $t = \tau + 1, \tau + 2, \ldots, T$ **do**
10 Observe X_t and compute MLE $\hat{\theta}_t \in \arg\max_{\theta \in \Theta} \sum_{s=1}^{t-1} l(\theta | Y_s, \{i_s, j_s\}, X_s)$;
11 Get $\mathbb{C}_t := \left\{\theta \in \mathbb{R}^d : \|\theta - \hat{\theta}_t\|_{M_t} \leq \beta_t\right\}$;
12 Observe safe context vectors $X_t^S = \left\{x \in X_t : \theta^T B x \leq c, \forall \theta \in \mathbb{C}_t\right\}$;
13 $\mathcal{A}_t = \{i \in \mathcal{A} : x_{t,i} \in X_t^S\}$;
14 Sample $B_t \sim \text{Ber}(p_t)$;
15 **if** $B_t = 1$ **then**
16 Sample $\tilde{\epsilon}_{t,i} \sim G$ for each $i \in \mathcal{A}_t$;
17 **else**
18 Sample $\tilde{\epsilon} \sim G$ and set $\tilde{\epsilon}_{t,i} = \tilde{\epsilon}$ for each $i \in \mathcal{A}_t$;
19 **end**
20 $\epsilon_{t,i} = \min(C_{\text{thresh}}, \max(-C_{\text{thresh}}, \tilde{\epsilon}_{t,i})) \quad \forall i \in \mathcal{A}_t$;
21 $i_t = \arg\max_{i \in \mathcal{A}_t} x_{t,i}^T \hat{\theta}_t + \epsilon_{t,i} \|x_{t,i}\|_{M_t^{-1}}$;
22 $j_t = \arg\max_{i \in \mathcal{A}_t} \langle z_{t,i_t,i}, \hat{\theta}_t \rangle + c_1 \|z_{t,i_t,i}\|_{M_t^{-1}}$;
23 Choose (i_t, j_t) and observe $Y_t = \mathbf{1}_{[i_t \succ j_t]}$;
24 Update $M_{t+1} \leftarrow M_t + z_{t,i_t,j_t} z_{t,i_t,j_t}^T$;
25 **end**

This phase plays a crucial role in the algorithm. It ensures that the pure exploration procedure collects sufficiently informative comparisons so that the unknown preference parameter θ^* becomes identifiable and the Gram matrix $M_{\tau+1}$ is well conditioned. As shown in Lemma B.1, under Assumptions 2.1 and 2.4, the exploration phase guarantees $\lambda_{\min}(M_{\tau+1}) \geq \frac{\lambda_- \tau}{2}$ with high probability. This spectral lower bound is essential for constructing reliable confidence regions and for safely expanding the admissible action set in subsequent rounds.

Safe Exploration–Exploitation Phase. Starting from round $t = \tau + 1$, the learner enters the safe exploration–exploitation phase (lines 8–25). At each round, given the history of observations $\{(Y_s, \{i_s, j_s\}, X_s)\}_{s=1}^{t-1}$, the learner computes the maximum likelihood estimate $\hat{\theta}_t$ (line 10), where the log-likelihood function for a single observation $(1_{[i_t \succ j_t]}, \{i_t, j_t\}, X_t)$ is given by

$$l(\theta) = 1_{[i_t \succ j_t]} \log F^*\big(\theta^\top (x_{t,i} - x_{t,j})\big) + \big(1 - 1_{[i_t \succ j_t]}\big) \log F^*\big(\theta^\top (x_{t,j} - x_{t,i})\big)\,,$$

and F^* denotes the comparison function induced by the perturbation distribution G^*.

Based on the maximum likelihood estimate $\hat{\theta}_t$ and the accumulated information matrix M_t, the learner constructs a confidence region $\mathbb{C}_t$ (line 11). The statistical validity of this confidence region is guaranteed by Lemma A.1, which is stated in Appendix A.

In particular, Lemma A.1 ensures that, under the stated assumptions and a sufficiently long initial exploration phase, the confidence region $\mathbb{C}_t$ contains the true parameter θ^* uniformly over all rounds $t > \tau$ with probability at least $1 - \delta$. This uniform containment property is a key ingredient in our regret analysis.

Leveraging this uncertainty quantification, the learner refines the admissible action space by constructing a time-dependent conservative safe set

$$X_t^S := \{x \in X_t : \theta^\top B x \le c,\ \forall \theta \in \mathbb{C}_t\}, \tag{7}$$

which contains only those actions that satisfy the safety constraint uniformly over the confidence region.

This safe decision set induces the feasible arm index set

$$\mathcal{A}_t := \{i \in \mathcal{A} : x_{t,i} \in X_t^S\}. \tag{8}$$

Consequently, any arm $i \in \mathcal{A}_t$ satisfies the safety constraint with high probability by construction.

In lines 14–20, the algorithm specifies the mechanism for generating and processing perturbations. First, in the perturbation generation step (lines 14–19), the learner samples a Bernoulli random variable $B_t \sim \mathrm{Ber}(p_t)$ to determine the exploration mode. If $B_t = 1$, independent perturbations $\tilde{\epsilon}_{t,i} \sim G$ are sampled for each arm i; otherwise, a single random variable $\tilde{\epsilon} \sim G$ is drawn and shared across all arms, i.e., $\tilde{\epsilon}_{t,i} = \tilde{\epsilon}$ for all i. Next, in the truncation step (line 20), each perturbation is clipped as

$$\epsilon_{t,i} = \min\big(C_{\mathrm{thresh}},\ \max(-C_{\mathrm{thresh}}, \tilde{\epsilon}_{t,i})\big),$$

which prevents excessively large perturbations from destabilizing the estimation process.

Following the CoLST framework [8], we adopt a utility maximization perspective for preference learning. At each round, the algorithm selects a pair of arms (i_t, j_t) as specified in lines 21–22.

Specifically, in line 21, the *first arm* is selected according to a perturbed optimistic principle:

$$i_t = \arg\max_{i \in \mathcal{A}_t} \Big(x_{t,i}^\top \hat{\theta}_t + \epsilon_{t,i} \, \|x_{t,i}\|_{M_t^{-1}} \Big),$$

where the perturbations $\epsilon_{t,i}$ are generated via a coupling mechanism controlled by the probability p_t, thereby interpolating between independent exploration across arms and shared randomness.

Given the selected first arm i_t, line 22 determines the *second arm* by maximizing an upper confidence bound over pairwise utility differences:

$$j_t = \arg\max_{i \in \mathcal{A}_t} \Big(\langle z_{t,i_t,i}, \hat{\theta}_t \rangle + c_1 \, \|z_{t,i_t,i}\|_{M_t^{-1}} \Big).$$

This choice promotes comparisons that are both statistically informative and aligned with optimistic utility estimates, a principle that has been shown to be effective in non-contextual dueling bandit and preference-based learning settings [7].

Finally, the learner observes the preference outcome $Y_t = 1_{[i_t \succ j_t]}$ (line 23) and updates the Gram matrix (line 24), which refines both the parameter estimate and the safe set in subsequent rounds.

4 Regret Analysis of Safe-COLSTIM

In our setting, the true safety set X_0^S is unknown because θ^* is unknown. Therefore, at each round t, the learner selects its action from the conservative safe inner approximation $X_t^S \subseteq X_0^S$. Intuitively, a better approximation leads to better cumulative regret, ideally matching the original COLSTIM rate $\mathcal{O}(d\sqrt{T}\log T)$.

At round t, the learner chooses a duel pair $S_t = (i_t, j_t)$. The optimal arm under the true safety set is

$$x_{t,i^*} = \arg\max_{x \in X_0^S} \theta^{*\top} x.$$

A key difference from classical COLSTIM is that the optimal arm x_{t,i^*} may not belong to the safe set at every round.

Based on the safe feasible arm set $\mathcal{A}_t$ defined in (8), the *safe optimistic arm* is given by

$$i_S^*(t) = \arg\max_{i \in \mathcal{A}_t} \langle x_{t,i}, \theta^* \rangle.$$

The corresponding safe optimistic pair is then $(\theta^*, x_{t,i_S^*(t)})$.

With this definition, we expand the instantaneous average regret into the following decomposition:

$$\begin{aligned} 2r_t^a(S_t, X_t) &= u^*_{i^*(t)}(X_t) - u^*_{i_t}(X_t) + u^*_{i^*(t)}(X_t) - u^*_{j_t}(X_t) \\ &= \underbrace{\left(\langle x_{t,i_S^*}, \theta^*\rangle - \langle x_{t,i_t}, \theta^*\rangle\right) + \left(\langle x_{t,i_S^*}, \theta^*\rangle - \langle x_{t,j_t}, \theta^*\rangle\right)}_{\text{Term I}} + 2\underbrace{\left(\langle x_{t,i^*}, \theta^*\rangle - \langle x_{t,i_S^*}, \theta^*\rangle\right)}_{\text{Term II}}. \end{aligned} \tag{9}$$

The goal is now to bound Term I and Term II separately. Term I resembles the regret component that appears in the classical COLSTIM analysis, while Term II captures the discrepancy between the true safe set X_0^S and its time-dependent estimate X_t^S.

Specifically, Term I reflects the regret incurred due to parameter estimation error and the gap between the true safe optimistic arm $i_S^*(t)$ and the selected arm pair $S_t = (i_t, j_t)$ under safety constraints. This type of regret is standard in preference-based and dueling bandit algorithms.

In contrast, Term II captures an additional *safe regret* arising from the safety constraints, and bounding it is inherently more subtle. In the absence of safety constraints, the corresponding Term II in classical COLSTIM is non-positive and therefore does not contribute to the regret. However, in Safe-COLSTIM, the optimal arm x_{t,i^*} may not belong to the estimated safe set X_t^S, which implies that Term II can be positive. This positive contribution represents the unavoidable cost of restricting actions to the safe set, and quantifies the price paid by Safe-COLSTIM for ensuring safety at every round.

4.1 Problem-Dependent Upper Bound

In this section, we upper bound the regret of Safe-COLSTIM through a problem-dependent quantity that we refer to as the *safety gap*, defined as

$$\Delta := c - \theta^{*\top} B x_{t,i^*} \geq 0.$$

The safety gap measures how far the optimal arm under the true parameter lies from the boundary of the safety constraint. When Δ is large, the optimal arm is well separated from the unsafe region, and small estimation errors in θ are unlikely to exclude it from the estimated safe set. Conversely, when Δ is close to zero, the optimal arm lies near the constraint boundary, making the learning problem inherently more delicate. As we show below, the magnitude of Δ directly determines how long the learner must explore before the estimated safe set reliably contains the true optimal arm, and thus plays a central role in controlling the safety-induced regret.

Lemma 4.1 ($x_{t,i^*} \in X_t^S$)**.** *Let Assumptions 2.1, 2.2, and 2.4 hold. Fix any $\delta \in (0,1)$ and assume a positive safety gap $\Delta > 0$. Initialize Safe-COLSTIM with $\tau \geq t_\delta := \frac{32}{\lambda_-}\log\left(\frac{d}{\delta}\right)$, where λ_- is defined in Lemma B.1. If*

$$\tau \geq T_\Delta := \left(\frac{8\|B\|^2\beta_T^2}{\lambda_-\,\Delta^2}\right) \vee t_\delta, \tag{10}$$

then, with probability at least $1-\delta$, it holds that $x_{t,i^*} \in X_t^S$ for all $t = \tau+1, \ldots, T$.

Lemma 4.1 implies that once the pure exploration phase is sufficiently long, the estimated safe set X_t^S contains the truly optimal safe arm at all later rounds. In this regime, the conservative nature of the safety constraint no longer excludes optimal actions, and the safety-induced regret ceases to accumulate.

In particular, under the condition $\Delta > 0$, the event $x_{t,i^*} \in X_t^S$ ensures that the safety-related component of the regret becomes non-positive. Equivalently, for all $t \geq \tau + 1$,

$$\text{Term II} := \langle x_{t,i^*}, \theta^*\rangle - \langle x_{t,i_S^*}, \theta^*\rangle \leq 0.$$

As a result, the regret of Safe-COLSTIM beyond the exploration phase is governed solely by the standard exploration– exploitation trade-off. The proof of Lemma 4.1 obtained by proving an equivalent reformulation given in Appendix D.1 (Lemma D.1).

Theorem 4.2 (Problem-dependent Regret bound; $\Delta > 0$)**.** *Suppose Assumptions 2.1, 2.2, 2.3, and 2.4 are satisfied, and the safety gap is strictly positive: $\Delta > 0$. Set the confidence parameter*

$$c_1 = \frac{2\sigma}{\mu^2}\sqrt{\frac{d}{2}\log\left(1+\frac{8T}{d}\right)+\log\left(\frac{1}{\delta}\right)}.$$

Define the threshold round

$$t' = \max\left\{t_\delta,\ \frac{8\sigma^2\left(\frac{d}{2}\log\left(1+\frac{8T}{d}\right)+\log\left(\frac{1}{\delta}\right)\right)}{\mu^2\lambda_-},\ \frac{2}{\lambda_-}\right\},$$

where μ represents the minimum slope of the comparison function F^*, and t_δ and λ_- are defined in Lemma B.1 via Assumptions 2.1 and 2.4. For any $c_2 \in (0, c_1]$, let the coupling probability be

$$p_t = \min\left(1, \frac{\sqrt{2d}}{2\sqrt{t-\tau}}(3c_1+c_2)\sqrt{\log\left(\frac{2T}{d}\right)}\right).$$

Run Safe-COLSTIM with exploration length $\tau \geq T_\Delta \vee t'$. Then, for any $C_{\text{thresh}} \in (0, c_2)$, the expected cumulative average regret satisfies

$$\mathbb{E}[R_T^a(\text{Safe-COLSTIM})] \leq 2\Delta_{\max}\tau + \Delta_{\max}(p_1+p_2)T + \Delta_{\max}\sum_{t=\tau+1}^{T} p_{3,t}$$
$$+\frac{c_3}{2}(3c_1+c_2)\sqrt{T\log\left(\frac{2T}{d}\right)} = \mathcal{O}\left(d\sqrt{T}\log T\right).$$

where $\Delta_{\max} = 2\sqrt{2}$ denotes the maximal average regret for the chosen arm pair.

Corollary 4.3 (Problem-dependent bound with spectral growth)**.** *Consider the setting of Theorem 4.2. If the Gram matrix sequence $\{M_t\}_{t=\tau+1}^T$ additionally satisfies the spectral growth property*

$$\sum_{t=\tau+1}^{T} \lambda_{\min}^{-1/2}(M_t) \ \leq\ b\sqrt{T}$$

for some absolute constant $b > 0$, where $\lambda_{\min}(M_t)$ is the smallest eigenvalue of M_t, then Safe-COLSTIM *achieves an improved expected cumulative average regret bound of $\mathbb{E}[R_T^a(\text{Safe-COLSTIM})] = \mathcal{O}\left(\sqrt{dT}\log T\right)$.*

The proofs of Theorem 4.2 and Corollary 4.3 are provided in Appendices E.3 and E.4, respectively.

4.2 Worst-Case Regret Bound

We now turn to the more challenging regime in which the safety gap Δ is zero or unavailable to the learner. In contrast to the problem-dependent setting, the absence of a strictly positive safety margin prevents us from guaranteeing that the true optimal safe arm eventually lies inside the estimated safe set within a logarithmic number of rounds. As a result, the safety-related regret term may persist throughout the learning process, and a refined analysis is required to establish sublinear regret guarantees.

Despite this difficulty, we show that Safe-COLSTIM still achieves vanishing average regret. In particular, by carefully balancing the duration of the initial pure exploration phase and the subsequent exploration-exploitation phase, the algorithm attains a worst-case regret rate of order $\mathcal{O}(T^{2/3}\log T)$. This bound holds uniformly when $\Delta = 0$ as well as when the safety gap is unknown, and therefore provides a general performance guarantee for Safe-COLSTIM in the absence of a favorable problem structure.

Theorem 4.4 (Worst-Case Regret Bound; $\Delta = 0$)**.** *Retain all conditions from Theorem 4.2, but now assume either $\Delta = 0$ or that the safety gap is unavailable to the learner. Run* Safe-COLSTIM *with*

exploration length $\tau \geq t'$ and coupling probabilities $\{p_t\}_{t\geq\tau+1}$ as previously specified. By setting $\tau = \left(\frac{\beta_T T}{\sqrt{\lambda_-}}\right)^{2/3} \vee t'$, the algorithm guarantees an expected cumulative average regret of

$$\mathbb{E}[R_T^a(\text{Safe-COLSTIM})] \leq 2\Delta_{\max}\tau + \Delta_{\max}(p_1+p_2)T + \Delta_{\max}\sum_{t=\tau+1}^{T} p_{3,t}$$
$$+\frac{2\sqrt{2}\,\beta_T\|B\|(T-\tau)}{c\sqrt{\lambda_-\,\tau}} + \frac{c_3}{2}(3c_1+c_2)\sqrt{T\log\left(\frac{2T}{d}\right)} = \mathcal{O}\left(dT^{2/3}\log T\right).$$

Corollary 4.5 (Worst-case bound with spectral growth)**.** *In the setting of Theorem 4.4, if the Gram matrix sequence $\{M_t\}_{t=\tau+1}^{T}$ satisfies the growth property $\sum_{t=\tau+1}^{T}\lambda_{\min}^{-1/2}(M_t) \leq b\sqrt{T}$, then* Safe-COLSTIM *achieves* $\mathbb{E}[R_T^a(\text{Safe-COLSTIM})] = \mathcal{O}\big((dT)^{2/3}\log T\big)$.

The proofs of Theorem 4.4 and Corollary 4.5 are provided in Appendices E.5 and E.6, respectively.

5 Experiments

In this section, we evaluate the performance of Safe-COLSTIM in safety-constrained contextual dueling bandit settings on both synthetic and real-world datasets, covering controlled environments and realistic application scenarios. Unless otherwise specified, the contextual dimension is set to $d = 10$ and the number of arms presented at each round is $n = 50$. The total number of rounds is $T = 10^4$. Preference feedback follows the Bradley–Terry–Luce (BTL) model, implemented by sampling perturbations from a standard Gumbel distribution. The initial pure exploration length is set to $\tau = t_0 = dn$, and we choose the threshold and confidence-related parameters as $c_1 = C_{\text{thresh}} = \eta = \sqrt{d\log T}$. The coupling probability is set to $p_t = \min\Big(1, \frac{d}{\sqrt{t-\tau}}\log(dT)\Big)$, which is a simplified version consistent with our theoretical analysis.

For the safety-constrained setting, we fix the confidence level $\delta = 0.01$ and the regularization parameter $\lambda = 1$. The safety matrix $B = \text{diag}(w) \in \mathbb{R}^{d\times d}$ is a diagonal matrix, where each entry w_i encodes the safety weight associated with the i-th feature dimension, with values drawn from $[0, 1]$ to reflect heterogeneous sensitivity across dimensions. The safety threshold c is drawn uniformly from $[0, 1]$, inducing a safety gap defined as $\Delta := c - \theta^{*\top}Bx^*$.

Synthetic dataset. We maintain a global arm pool of size 2000. At each round t, the contextual decision set X_t is obtained by sampling $n = 50$ arms uniformly without replacement from this pool. Both the unknown parameter θ^* and the arm features in the pool are sampled uniformly from the unit ℓ_2-ball in $\mathbb{R}^d$. Results on the synthetic dataset are averaged over 20 independent runs.

Real-world datasets. We further evaluate Safe-COLSTIM on three real-world recommendation datasets:

- **Last.FM**: a music artist recommendation dataset containing 186,479 interactions between 1,892 users and 17,632 artists.
- **Movielens**: an extension of the MovieLens10M dataset augmented with IMDb and Rotten Tomatoes tag information, containing 47,957 interactions between 2,113 users and 10,197 movies.
- **Yelp**: a public dataset containing users' reviews of restaurants on Yelp. We retain only users with at least 100 reviews and restaurants with at least 50 reviews, yielding 342,237 reviews from 1,998 users on 25,160 restaurants.

For each real-world dataset, we follow a standard preprocessing pipeline [27]. We first select the $|\mathcal{A}| = 2000$ items with the most assigned attributes and the $N_u = 20$ users with the most assigned attributes. For each arm, we construct a high-dimensional feature vector from its full attribute information, and then apply PCA to reduce the dimensionality, retaining the top 10 principal components as the contextual representation ($d = 10$). For Last.FM and MovieLens, the true preference feedback is inferred from interaction records: if a user has assigned attributes to an

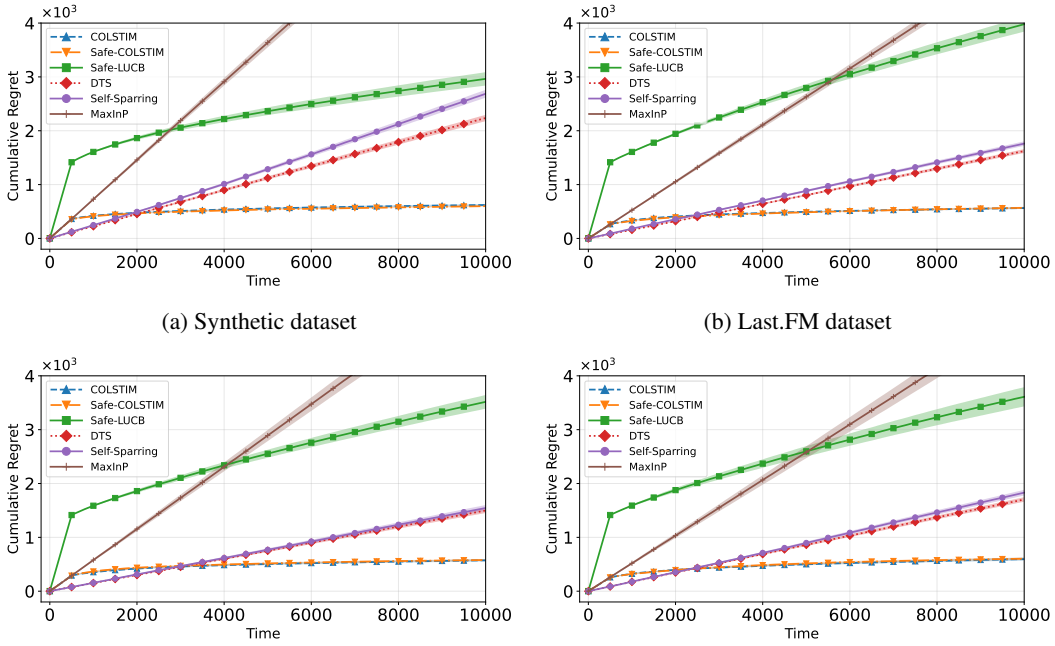
item, the feedback is 1; otherwise it is treated as missing. For Yelp, the original 5-level ratings are binarized—high ratings (4 and 5) are mapped to 1 and low ratings (≤ 3) to 0. The true preference parameter θ^* for each user is then estimated via ridge regression. At each round, we sample $n = 50$ arms uniformly without replacement from $\mathcal{A}$ to form the decision set $\tilde{\mathcal{A}}_t$, which is shared across all algorithms. For the real-world experiments, we do not average over repeated runs; instead, results are computed across the N_u users, which provides natural statistical variability.

All experiments are conducted on an AMD Ryzen 5 7500F CPU with 32 GB RAM.

Implementation of the seed set. To satisfy Assumption 2.4, we construct the seed set $\mathcal{S}_0 = \{r_0 e_j\}_{j=1}^d$ as $d = 10$ scaled standard basis vectors, where e_j denotes the j-th standard basis vector in $\mathbb{R}^d$ and $r_0 = c/(S \cdot \max_j w_j)$. Since $B = \mathrm{diag}(w)$, we have $Be_j = w_j e_j$, and thus $\max_{\theta \in \Theta} \theta^\top B(r_0 e_j) = S \cdot r_0 |w_j| \leq S \cdot r_0 \cdot \max_j w_j = c$, so each seed satisfies Assumption 2.4(i). The spanning condition $\frac{1}{|\mathcal{S}_0|} \sum_{j=1}^d (r_0 e_j)(r_0 e_j)^\top = \frac{r_0^2}{d} I_d$ verifies Assumption 2.4(ii) with $\rho_0 = r_0^2/d$. At each round, the seed set is included in X_t alongside $n - d = 40$ arms freshly sampled (from the global pool for the synthetic dataset, or from $\mathcal{A}$ for real-world datasets) without replacement, satisfying Assumption 2.4(iii). During the pure exploration phase, the algorithm forms $X^w = \{x \in X_t : \max_{\theta \in \Theta} \theta^\top Bx \leq c\} \cup \mathcal{S}_0$ and samples arm pairs uniformly without replacement from this set.

5.1 Overall Performance and Safety Evaluation

Our first experiment focuses on the overall performance comparison and safety evaluation. Under the unified experimental setting described above, we compare Safe-COLSTIM with the non-safety-constrained contextual dueling bandit algorithms COLSTIM [8] and MaxInP [19], the safety-constrained linear bandit algorithm Safe-LUCB [4], and two non-contextual dueling bandit algorithms, DTS [25] and Self-Sparring [22]. We evaluate the learning efficiency of these algorithms by their cumulative average regret, and additionally record the number and probability of safety violations to quantify the effectiveness of the safety mechanism.



(a) Synthetic dataset

(b) Last.FM dataset

(c) Movielens dataset

(d) Yelp dataset

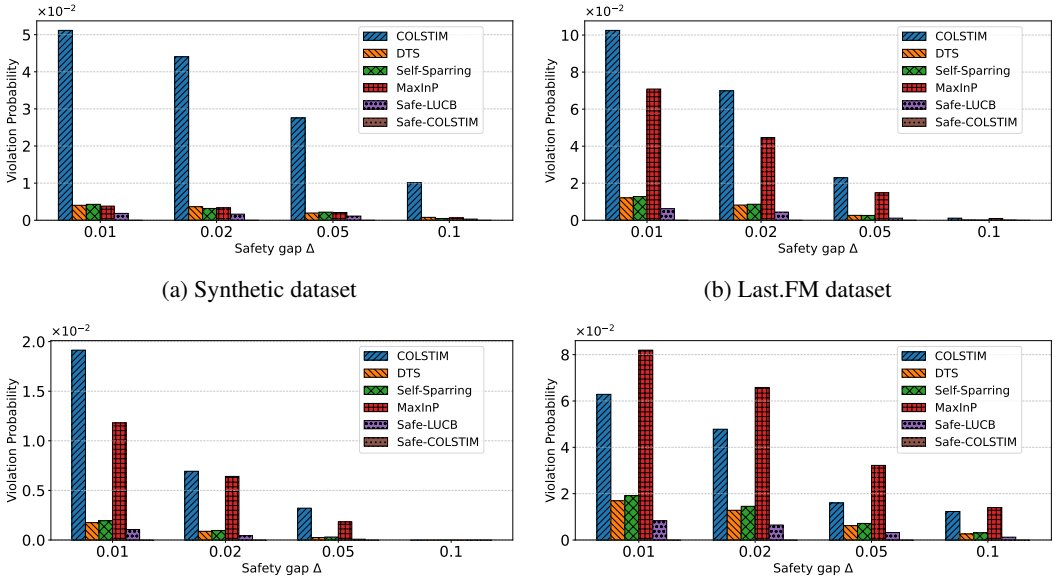
Figure 1: Cumulative regret on synthetic and real-world datasets.

Cumulative regret. Fig. 1 reports the cumulative regret curves on the synthetic dataset and the three real-world datasets. The relative ordering of the algorithms is highly consistent across all four

datasets, which suggests that the observed comparisons are driven by algorithmic design rather than dataset-specific artifacts.

Among the baselines, MaxInP exhibits the worst performance, with a nearly linear regret growth on every dataset. This is because MaxInP actively selects highly uncertain but potentially suboptimal arm pairs to maximize information gain, which is effective for narrowing the confidence region but incurs substantial regret per round. Safe-LUCB displays a characteristic pattern: a large initial regret jump during its conservative warm-up phase, followed by approximately linear but slower growth. Although Safe-LUCB enforces safety, it is fundamentally designed for linear bandits with numerical rewards and is not tailored to pairwise preference feedback, which limits its asymptotic efficiency. DTS and Self-Sparring perform similarly to each other and fall in between, achieving lower regret than MaxInP and Safe-LUCB but still showing roughly linear growth, since they neither leverage contextual information nor employ a structured pure exploration phase.

In contrast, Safe-COLSTIM and COLSTIM achieve the lowest cumulative regret across all four datasets, with their curves nearly overlapping and rapidly flattening after the initial exploration phase. This near-coincidence is the key empirical finding: the safety mechanism in Safe-COLSTIM introduces only a negligible regret overhead relative to its unconstrained counterpart, confirming that strict per-round safety constraints can be enforced without sacrificing learning efficiency.



(a) Synthetic dataset

(b) Last.FM dataset

(c) Movielens dataset

(d) Yelp dataset

Figure 2: Average safety violation probability under different safety gaps Δ.

Safety violation evaluation. We further evaluate the safety performance of different algorithms by reporting the average number of safety violations and the corresponding per-round violation probability, defined as Avg Violations$/T$. For this evaluation, we use $T = 5000$ rounds to better observe violation patterns. Table 1 summarizes these metrics under four representative safety gap values $\Delta \in \{0.01, 0.02, 0.05, 0.10\}$, and Fig. 2 visualizes the resulting violation probabilities.

Several consistent trends emerge from these results. First, the safety violation probability of all unsafe baselines decreases as Δ increases, since a larger safety gap relaxes the constraint and expands the set of feasible arms. Second, non-safety-aware bandit algorithms incur substantially higher violation probabilities across all Δ values. In particular, COLSTIM and MaxInP suffer the highest violation rates, with up to 10.6% per-round violation on Last.FM and 8.5% on Yelp when $\Delta = 0.01$, because their optimism-driven or information-driven selection rules actively push arms toward the boundary of the feasible region. DTS and Self-Sparring exhibit lower violation rates than COLSTIM, but this is largely a byproduct of their inefficient exploration rather than any safety-aware design. Safe-LUCB reduces violations relative to non-safe baselines but still incurs noticeable violations across all datasets (e.g., 1.07% on Yelp at $\Delta = 0.01$), because its safety mechanism is designed for single-arm linear

Table 1: Violation statistics across four datasets (Avg = average number of violations over $T = 5000$ rounds; Prob. = per-round violation probability). Safe-COLSTIM consistently achieves zero safety violations across all safety gap settings and all datasets.

		Synthetic		Last.FM		MovieLens		Yelp	
Δ	Algorithm	Avg	Prob.	Avg	Prob.	Avg	Prob.	Avg	Prob.
0.01	COLSTIM	312.85	0.0626	529.70	0.1059	102.85	0.0206	343.00	0.0686
	DTS	24.95	0.0050	70.55	0.0141	10.85	0.0022	104.50	0.0209
	Self-Sparring	20.00	0.0040	59.85	0.0120	9.35	0.0019	97.75	0.0196
	MaxInP	24.95	0.0050	394.75	0.0789	67.75	0.0136	425.45	0.0851
	Safe-LUCB	12.20	0.0024	38.30	0.0077	6.00	0.0012	53.50	0.0107
	Safe-COLSTIM	**0.00**	**0.0000**	**0.00**	**0.0000**	**0.00**	**0.0000**	**0.00**	**0.0000**
0.02	COLSTIM	233.70	0.0467	371.55	0.0743	41.05	0.0082	252.10	0.0504
	DTS	22.05	0.0044	51.60	0.0103	4.45	0.0009	75.60	0.0151
	Self-Sparring	16.55	0.0033	46.30	0.0093	3.45	0.0007	70.05	0.0140
	MaxInP	20.20	0.0040	272.40	0.0545	32.25	0.0065	329.30	0.0659
	Safe-LUCB	11.30	0.0023	27.15	0.0054	1.65	0.0003	41.20	0.0082
	Safe-COLSTIM	**0.00**	**0.0000**	**0.00**	**0.0000**	**0.00**	**0.0000**	**0.00**	**0.0000**
0.05	COLSTIM	169.55	0.0339	163.05	0.0326	18.35	0.0037	106.95	0.0214
	DTS	12.75	0.0026	15.55	0.0031	1.10	0.0002	37.65	0.0075
	Self-Sparring	7.95	0.0016	11.40	0.0023	1.30	0.0003	38.15	0.0076
	MaxInP	12.90	0.0026	72.10	0.0144	13.60	0.0027	175.60	0.0351
	Safe-LUCB	7.05	0.0014	7.50	0.0015	0.45	0.0001	20.20	0.0040
	Safe-COLSTIM	**0.00**	**0.0000**	**0.00**	**0.0000**	**0.00**	**0.0000**	**0.00**	**0.0000**
0.10	COLSTIM	61.40	0.0123	6.25	0.0013	0.00	0.0000	66.00	0.0132
	DTS	3.90	0.0008	0.40	0.0001	0.00	0.0000	16.05	0.0032
	Self-Sparring	4.10	0.0008	0.50	0.0001	0.00	0.0000	16.20	0.0032
	MaxInP	3.95	0.0008	4.30	0.0009	0.00	0.0000	70.15	0.0140
	Safe-LUCB	1.90	0.0004	0.45	0.0001	0.00	0.0000	7.35	0.0015
	Safe-COLSTIM	**0.00**	**0.0000**	**0.00**	**0.0000**	**0.00**	**0.0000**	**0.00**	**0.0000**

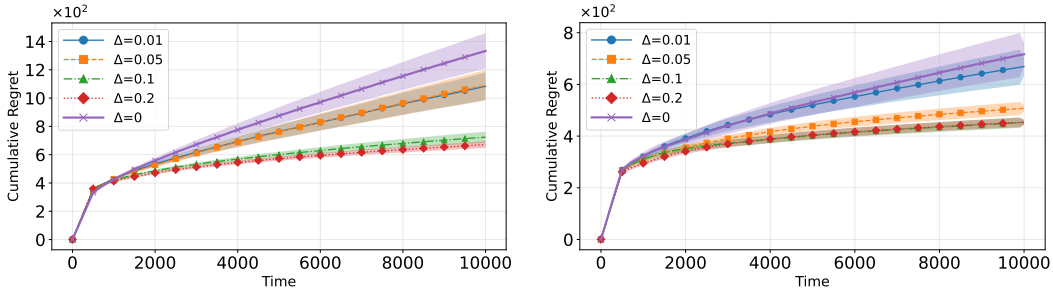
bandits and can certify safety for only one of the two arms in a duel. In stark contrast, Safe-COLSTIM consistently achieves *exactly zero* safety violations across all four datasets and across every Δ setting, including the most stringent $\Delta = 0.01$ case where every other algorithm violates the constraint hundreds of times. Combined with the regret results above, this demonstrates that Safe-COLSTIM is the only method among those considered that simultaneously attains competitive learning efficiency and strict per-round safety.

5.2 Ablation Study

We next conduct an ablation study to isolate the effects of several key components in Safe-COLSTIM, including the safety gap Δ, the safety constraint mechanism, and the pure exploration phase length τ. Each experiment modifies a single component while keeping all other settings fixed.

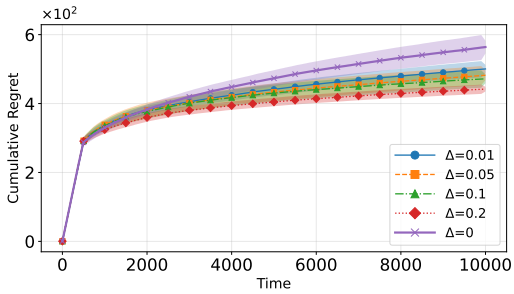
Effect of the safety gap Δ. Fig. 3 studies how the safety gap Δ affects the cumulative regret of Safe-COLSTIM, with Δ ranging over $\{0, 0.01, 0.05, 0.1, 0.2\}$. Across all four datasets, the case $\Delta = 0$ consistently incurs the highest cumulative regret, and the gap between $\Delta = 0$ and the strictly positive cases grows steadily over time. This behavior is in agreement with our theoretical result, which predicts a worse $\mathcal{O}(dT^{2/3} \log T)$ rate when $\Delta = 0$ compared to the $\mathcal{O}(d\sqrt{T \log T})$ rate obtained under a strictly positive safety gap. Among the cases with $\Delta > 0$, the cumulative regret generally decreases as Δ increases, since a larger safety gap induces a looser estimated safe set and allows the learner to explore a richer set of candidate arms. This trend is most visible on the synthetic dataset, where the curves for different $\Delta > 0$ values remain clearly separated throughout the horizon. On the three real-world datasets, the curves for $\Delta \in \{0.05, 0.1, 0.2\}$ become close to each other once Δ is sufficiently large, suggesting that the marginal benefit of further increasing Δ diminishes once the estimated safe set is no longer the binding constraint on exploration.

Effect of the safety constraint mechanism. Fig. 4 compares Safe-COLSTIM with its unconstrained counterpart under two regimes, $\Delta > 0$ and $\Delta = 0$, with all other settings held fixed. When $\Delta > 0$, the safety-aware curve and the unconstrained curve nearly overlap on all four datasets throughout the entire horizon, indicating that the proposed safety mechanism introduces essentially no additional

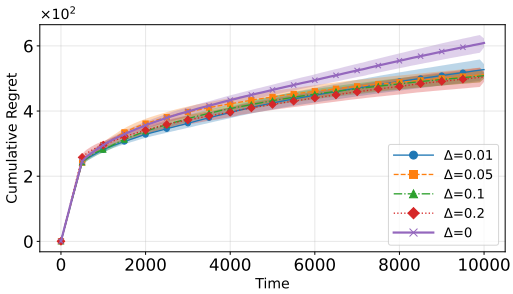


(a) Synthetic dataset

(b) Last.FM dataset



(c) Movielens dataset



(d) Yelp dataset

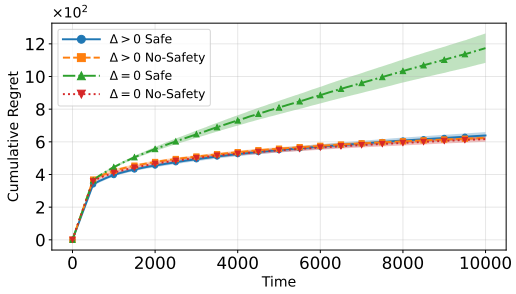
Figure 3: Cumulative regret of Safe-COLSTIM under different safety gaps Δ.

regret in this regime. When $\Delta = 0$, however, the safety-aware variant exhibits a noticeably larger regret than its unconstrained counterpart, with the gap widening over time. This separation is most pronounced on the synthetic and Last.FM datasets, and somewhat milder on MovieLens and Yelp, but is consistently visible across all four. The pattern reflects an unavoidable cost of enforcing safety when the optimal arm lies on the boundary of the feasible region: the conservative safe set repeatedly excludes near-optimal arms, leading to the $\mathcal{O}(dT^{2/3}\log T)$ regret scaling established in Theorem 4.4. Crucially, this additional regret is the price paid for eliminating safety violations entirely (Section 5.1), and remains sublinear in T.

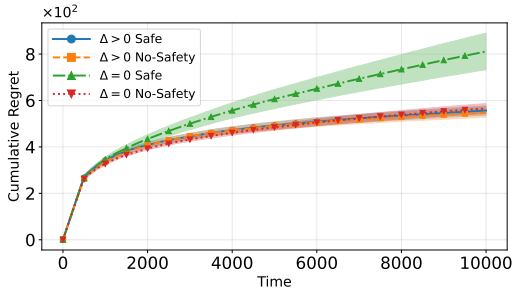
Effect of the pure exploration phase length τ. Fig. 5 examines the role of the pure exploration phase by varying its length over $\tau \in \{0, \sqrt{T}, dn, T^{2/3}\}$. Across all four datasets, $\tau = \sqrt{T}$ consistently achieves the lowest cumulative regret over the full horizon, while $\tau = T^{2/3}$ yields the highest regret on the synthetic and MovieLens datasets due to the large linear cost paid during a prolonged pure exploration phase. The choice $\tau = 0$ exhibits a distinctive shape across all four datasets: it accumulates the least regret in the very early rounds since no exploration cost is paid, but its regret subsequently grows faster than the other curves and eventually catches up to or surpasses them in later rounds. This pattern is most pronounced on Last.FM, where $\tau = 0$ ends with the largest regret, and is also visible—though less dramatically—on the synthetic, MovieLens, and Yelp datasets. The underlying reason is that without an initial exploration phase, the Gram matrix is poorly conditioned and the preference parameter estimate is unreliable, leading to persistent suboptimal selections whose cost accumulates over time. The intermediate choice $\tau = dn$ falls between $\tau = \sqrt{T}$ and the two extremes on most datasets. On the Yelp dataset, all four choices produce closely overlapping curves, suggesting that the regret is less sensitive to τ in this setting. Overall, these results highlight the need to balance early information acquisition against the linear cost of pure exploration, and confirm that a moderate τ is essential to the performance of Safe-COLSTIM.

6 Related Work

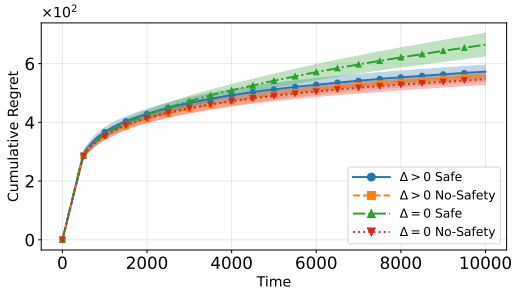
The contextualized extension of preference-based bandits with pairwise comparisons has received limited attention. Dudík et al. [14] first studied this setting by assuming access to a space of learning policies, though their framework differs from ours. Saha [19] considered a latent linear



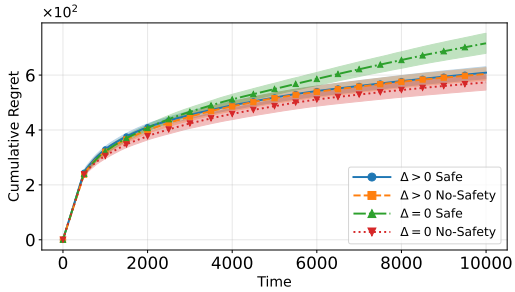
(a) Synthetic dataset



(b) Last.FM dataset

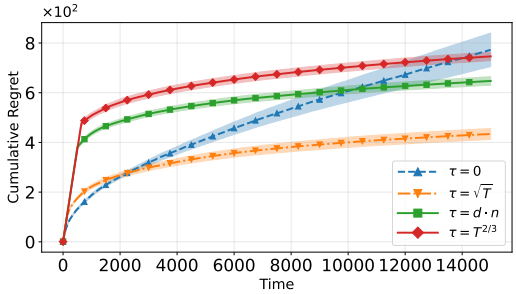


(c) Movielens dataset

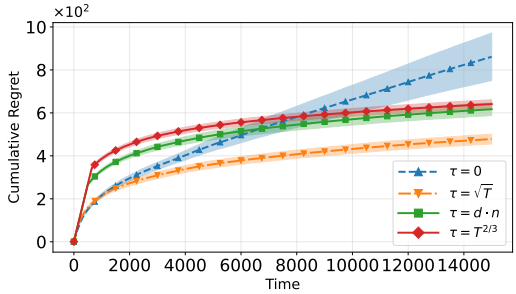


(d) Yelp dataset

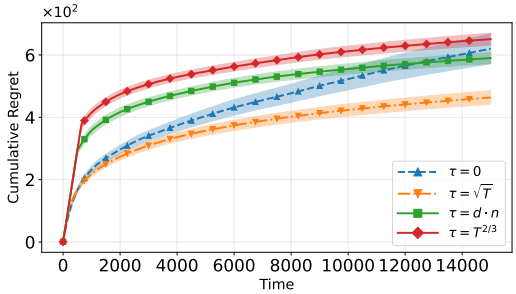
Figure 4: Impact of the safety constraint mechanism on cumulative regret.



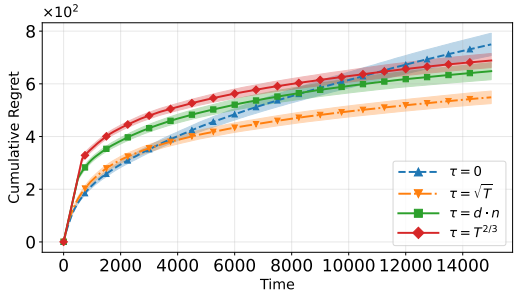
(a) Synthetic dataset



(b) Last.FM dataset



(c) Movielens dataset



(d) Yelp dataset

Figure 5: Effect of the pure exploration phase length τ on cumulative regret.

utility model with a logit link function, corresponding to the contextualized Bradley-Terry-Luce model, and established a lower bound of $\Omega(\sqrt{dT})$. More closely related, Bengs et al. [8] introduced the Contextual Linear Stochastic Transitivity (CoLST) model and employed randomized learning strategies inspired by [24]. Such randomized approaches offer computational advantages by requiring only a sampling distribution rather than closed-form posteriors. Our work extends the CoLST framework by incorporating safety constraints.

In linear stochastic bandits, UCB-based algorithms [5, 13, 1] have achieved $O(\sqrt{T \log T})$ regret for unconstrained settings. Two main constraint frameworks have emerged: global budget constraints limiting resource consumption [6, 3], and stage-wise safety maintaining performance relative to baselines [16, 26]. These constraints apply cumulatively and depend on the algorithm history. More relevant to our work, Amani et al. [4] studied linear stochastic bandits under per-round linear safety constraints, proposing Safe-LUCB with conservative confidence sets for high-probability safety. Beyond bandits, safe exploration has been studied in reinforcement learning through constrained policy search [2] and Gaussian process-based frameworks [21, 9]. To our knowledge, our work is the first to integrate per-round linear safety constraints into contextual dueling bandits, addressing the challenge of balancing preference-based exploration with strict safety enforcement under pairwise feedback.

7 Conclusion and Future Work

We studied the problem of safety-constrained contextual dueling bandits and proposed Safe-COLSTIM, which integrates conservative safety mechanisms with optimism-driven exploration. Theoretically, we established that under a positive safety gap $\Delta > 0$, the algorithm attains $O(d\sqrt{T \log T})$ regret, matching unconstrained methods, while in the worst case with $\Delta = 0$, it ensures $O((dT)^{2/3} \log T)$ regret. Our experiments validate these findings, showing near-zero safety violations across different settings while maintaining competitive performance. The ablation studies confirm the necessity of both the safety mechanism and the appropriate exploration phase.

This work opens several promising avenues for future research. First, integrating large language models into our framework presents exciting opportunities. An LLM could serve as a simulated user with consistent preference patterns, while our algorithm acts as a query model that learns optimal strategies through safe exploration, enabling safer model alignment under explicit constraint satisfaction. Second, extending our approach to more expressive preference structures—such as partial rankings or list-wise comparisons [10, 20] would broaden its applicability to recommendation systems and information retrieval. Additionally, relaxing the linearity assumption via kernel methods [18] or developing adaptive safety mechanisms could substantially enhance flexibility while preserving safety guarantees.

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A Self-Normalized Confidence Region for the MLE

This section provides a high-probability confidence region for the maximum-likelihood estimator $\hat{\theta}_t$, which is used as a technical ingredient in the regret analysis of the main text. The result relies on self-normalized martingale concentration techniques adapted to the dueling bandit setting, and holds provided that the initial pure exploration phase is sufficiently long to ensure an initial concentration event.

Lemma A.1 (Confidence Region for the MLE)**.** *Under Assumptions 2.1, and 2.3, suppose the initial exploration length τ is chosen such that initial concentration event*

$$A_{\text{init}} := \Big\{\forall t > \tau : \ \|\hat{\theta}_t - \theta^*\| \le 1 \text{ and } M_t \succeq I_d\Big\}$$

holds. Then for any $\delta > 0$, with probability at least $1 - \delta$ and for all $t > \tau$,

$$\|\hat{\theta}_t - \theta^*\|_{M_t} \ \le \ \beta_t := \frac{2\sigma}{\mu}\sqrt{\frac{d}{2}\log\Big(1 + \frac{8t}{d}\Big) + \log\frac{1}{\delta}}, \tag{11}$$

where $\sigma = 1$ is the sub-Gaussian parameter of the centered binary observations, which follows from Hoeffding's lemma since $Y_t \in \{0, 1\}$ has bounded support.

Proof. We begin by introducing auxiliary functions that will facilitate our analysis. Define the gradient discrepancy and score process:

$$G_t(\theta) := \sum_{s=1}^{t-1} \big(F(\langle Z_s, \theta\rangle) - F(\langle Z_s, \theta^*\rangle)\big)\, Z_s, \qquad V_t := \sum_{s=1}^{t-1} \epsilon_s Z_s.$$

First-order optimality. The MLE satisfies the first-order condition

$$\sum_{s=1}^{t-1} \big(Y_s - F(\langle Z_s, \hat{\theta}_t\rangle)\big) Z_s = 0.$$

Since $Y_s = F(\langle Z_s, \theta^*\rangle) + \epsilon_s$, rearranging yields

$$\sum_{s=1}^{t-1} \big(F(\langle Z_s, \hat{\theta}_t\rangle) - F(\langle Z_s, \theta^*\rangle)\big)\, Z_s = \sum_{s=1}^{t-1} \epsilon_s Z_s,$$

which can be compactly written as

$$G_t(\hat{\theta}_t) = V_t. \tag{12}$$

Strong monotonicity. We now establish a key inequality relating the gradient discrepancy to the parameter error. By the mean value theorem, for each s,

$$F(\langle Z_s, \theta\rangle) - F(\langle Z_s, \theta^*\rangle) = F'(\xi_s)\, Z_s^\top(\theta - \theta^*),$$

where ξ_s lies between $Z_s^\top\theta$ and $Z_s^\top\theta^*$. The regularity assumption ensures $F'(\xi_s) \ge \mu$, yielding

$$G_t(\theta) = \sum_{s=1}^{t-1} F'(\xi_s)\, Z_s Z_s^\top(\theta - \theta^*) \ \ge \ \mu\, M_t(\theta - \theta^*).$$

Consequently,

$$\|G_t(\theta)\|^2_{M_t^{-1}} \ge \mu^2(\theta - \theta^*)^\top M_t^\top M_t^{-1} M_t(\theta - \theta^*) = \mu^2\|\theta - \theta^*\|^2_{M_t}. \tag{13}$$

Combining optimality and monotonicity. Evaluating (13) at $\theta = \hat{\theta}_t$ and applying (12) gives

$$\|\hat{\theta}_t - \theta^*\|_{M_t} \ \le \ \frac{1}{\mu}\|V_t\|_{M_t^{-1}}. \tag{14}$$

Concentration of the score process. To complete the proof, we invoke Lemma A.2, which provides a self-normalized concentration bound. Specifically, with probability at least $1 - \delta$,

$$\|V_t\|^2_{M_t^{-1}} \ \le \ 4\sigma^2\Big(\frac{d}{2}\log\big(1 + \frac{8t}{d}\big) + \log\frac{1}{\delta}\Big). \tag{15}$$

Combining (14) with (15) and taking square roots yields the desired confidence radius. □

A.1 Self-Normalized Concentration Analysis

The preceding lemma relies on a self-normalized concentration result for the score process, which we now establish. This bound is crucial for controlling the uncertainty in our parameter estimate without requiring strong assumptions on the design sequence.

Lemma A.2 (Self-Normalized Concentration for the Score Process)**.** *Suppose there exists an integer τ such that $\lambda_{\min}(M_\tau) \geq 1$. Then for any $\delta > 0$, with probability at least $1 - \delta$, the following holds for all $t > \tau$:*

$$\|V_t\|^2_{M_t^{-1}} \;\leq\; 4\sigma^2 \left(\frac{d}{2} \log\left(1 + \frac{8t}{d}\right) + \log\frac{1}{\delta} \right),$$

where $V_t = \sum_{s=1}^{t-1} \epsilon_s Z_s$.

Proof. Fix $t > \tau$ and write $M = M_t$, $V = V_t$ for notational simplicity. We employ a regularization technique to facilitate the analysis.

Regularization decomposition. Define the regularized matrix $\overline{M} := M + \lambda I$ with $\lambda = \frac{1}{2}$. The self-normalized term admits the decomposition

$$\|V\|^2_{M^{-1}} = \|V\|^2_{\overline{M}^{-1}} + V^\top (M^{-1} - \overline{M}^{-1}) V. \tag{16}$$

Bounding the regularization term. We first bound the second term. By the Sherman–Morrison formula $A^{-1} = \overline{A}^{-1} - \frac{\alpha\, \overline{A}^{-1} u u^\top \overline{A}^{-1}}{1 + \alpha\, u^\top \overline{A}^{-1} u}$ with $\overline{A} = A + \alpha u u^\top$, applying it to $\overline{M} = M + \lambda I$ yields

$$\overline{M}^{-1} = M^{-1} - \frac{\lambda M^{-2}}{1 + \lambda\, \mathbf{1}^\top M^{-1} \mathbf{1}}.$$

Since $\mathbf{1}^\top M^{-1} \mathbf{1} \geq 0$, we obtain

$$0 \;\leq\; V^\top (M^{-1} - \overline{M}^{-1}) V = \frac{\lambda\, V^\top M^{-2} V}{1 + \lambda\, \mathbf{1}^\top M^{-1} \mathbf{1}} \;\leq\; \lambda\, V^\top M^{-2} V. \tag{17}$$

Using $\|M^{-1}\| = \lambda_{\max}(M^{-1}) = 1/\lambda_{\min}(M)$ and the assumption $\lambda_{\min}(M) \geq \lambda_{\min}(M_\tau) \geq 1$, we further derive

$$V^\top M^{-2} V \leq \|M^{-1}\|\, \|V\|^2_{M^{-1}} \leq \|V\|^2_{M^{-1}}. \tag{18}$$

Combining (17) with (18) gives

$$0 \;\leq\; V^\top (M^{-1} - \overline{M}^{-1}) V \;\leq\; \lambda\, \|V\|^2_{M^{-1}}. \tag{19}$$

The above inequality together with (16) implies that

$$\|V\|^2_{M^{-1}} \leq (1 - \lambda)^{-1}\, \|V\|^2_{\overline{M}^{-1}}.$$

Choosing $\lambda = \frac{1}{2}$ yields

$$\|V\|^2_{M^{-1}} \leq 2\, \|V\|^2_{\overline{M}^{-1}} = 2 \left\| \sum_{s=1}^{t-1} \epsilon_s Z_s \right\|^2_{\overline{M}^{-1}}. \tag{20}$$

Martingale concentration. We now bound the regularized norm using a self-normalized martingale inequality. By Lemma A.3, adapted from the analysis of [1], for all $t \geq 0$, with probability at least $1 - \delta$,

$$\|V_t\|^2_{\overline{M}^{-1}} \leq 2\sigma^2\, \log\left(\frac{\det(\overline{M})^{1/2}}{\det(\lambda I)^{1/2}\, \delta} \right). \tag{21}$$

Determinant bound. To bound the determinant ratio, we invoke Lemma A.4, which gives

$$\frac{\det(\overline{M})}{\det(\lambda I)} \;\leq\; \left(1 + \frac{tL^2}{d\lambda}\right)^d, \qquad L = 2. \tag{22}$$

where we used $\|Z_s\|_2 \leq 2$ and $\lambda = \frac{1}{2}$. Combining (21) and (22) yields

$$\|V_t\|^2_{\overline{M}^{-1}} \leq 2\sigma^2 \left(\frac{d}{2} \log\left(1 + \frac{8t}{d}\right) + \log\frac{1}{\delta} \right).$$

The claim follows by applying (20). □

A.2 Supporting Technical Lemmas

We conclude this section by stating the auxiliary results used in the preceding proofs. These lemmas provide foundational concentration and determinant inequalities.

Lemma A.3 (Self-Normalized Martingale Bound)**.** *Let $\{\mathcal{F}_t\}_{t=0}^{\infty}$ be a filtration. Let $\{Z_t\}_{t=1}^{\infty} \subset \mathbb{R}^d$ be a sequence of $\mathcal{F}_{t-1}$-measurable random vectors, and let $\{\epsilon_t\}_{t=1}^{\infty}$ be a sequence of real-valued random variables such that $\mathbb{E}[\epsilon_t \mid \mathcal{F}_{t-1}] = 0$ and ϵ_t is conditionally σ-sub-Gaussian, i.e.,*

$$\mathbb{E}[\exp(\lambda \epsilon_t) \mid \mathcal{F}_{t-1}] \leq \exp\left(\frac{\lambda^2 \sigma^2}{2}\right), \qquad \forall \lambda \in \mathbb{R}.$$

For $V_t = \sum_{s=1}^{t-1} \epsilon_s Z_s$ and $\overline{M} = \lambda I + M_t$, with probability at least $1-\delta$,

$$\|V_t\|^2_{\overline{M}^{-1}} \leq 2\sigma^2 \log\left(\frac{\det(\overline{M})^{1/2}}{\det(\lambda I)^{1/2}\,\delta}\right).$$

Proof. This follows from a standard application of the method of mixtures for self-normalized processes; see [1] for details. □

Lemma A.4 (Determinant-Trace Inequality)**.** *Let $Z_1, \ldots, Z_t \in \mathbb{R}^d$ satisfy $\|Z_s\|_2 \leq L$ for all s, and define $\overline{M} := \lambda I + \sum_{s=1}^{t} Z_s Z_s^\top$. Then $\det(\overline{M}) \;\leq\; \left(\lambda + \frac{tL^2}{d}\right)^d$.*

Proof. Let $\alpha_1, \ldots, \alpha_d > 0$ denote the eigenvalues of $\overline{M}$. By the arithmetic-geometric mean inequality,

$$\det(\overline{M})^{1/d} = \left(\prod_{i=1}^{d} \alpha_i\right)^{1/d} \leq \frac{1}{d}\sum_{i=1}^{d} \alpha_i = \frac{\operatorname{tr}(\overline{M})}{d}.$$

The trace satisfies

$$\operatorname{tr}(\overline{M}) = d\lambda + \sum_{s=1}^{t} \operatorname{tr}(Z_s Z_s^\top) = d\lambda + \sum_{s=1}^{t} \|Z_s\|_2^2 \leq d\lambda + tL^2.$$

Raising both sides to the d-th power completes the proof. □

B Learning the Safe Set

The main challenge in controlling the safety-induced regret is that the estimated safe set X_t^S generally differs from the true safe set X_0^S. At a high level, we relate $\mathrm{Term\ II}$ in the regret decomposition to a notion of *distance* between X_t^S and X_0^S measured in the direction of the optimal arm x_{t,i^*}. We then show that this distance can be controlled by lower bounding the minimum eigenvalue $\lambda_{\min}(M_t)$ of the Gram matrix. Since directly establishing such a bound for each round is challenging due to the interdependence of actions, we instead exploit the monotonicity $\lambda_{\min}(M_t) \geq \lambda_{\min}(M_{\tau+1})$ for all $t \geq \tau + 1$. The pure exploration phase of Safe-COLSTIM ensures a high-probability lower bound on $\lambda_{\min}(M_{\tau+1})$, implying that X_t^S is a sufficiently accurate approximation of X_0^S during the subsequent exploration–exploitation phase.

Lemma B.1. *Let $M_{\tau+1} = \sum_{t=1}^{\tau} z_t z_t^\top$ be the Gram matrix corresponding to the first τ rounds (pure exploration phase) of Safe-COLSTIM. Define*

$$\lambda_- \;:=\; \lambda_{\min}(\Sigma) \;=\; \lambda_{\min}\Big(\mathbb{E}_{X_t \sim \mathcal{D}}\, \mathbb{E}_{(x,y)\sim \pi_{\exp}(X_t)}\big[(x-y)(x-y)^\top\big]\Big),$$

where (x, y) is sampled uniformly without replacement from X^w as in line 4 of Algorithm 1. Under Assumptions 2.1 and 2.4, $\lambda_- > 0$. Moreover, for any $\delta \in (0,1)$, with probability at least $1-\delta$,

$$\lambda_{\min}(M_{\tau+1}) \;\geq\; \frac{\lambda_- \tau}{2},$$

provided that

$$\tau \;\geq\; t_\delta \;:=\; \frac{32}{\lambda_-}\log\Big(\frac{d}{\delta}\Big).$$

Proof. We proceed in two steps. We first establish a positive lower bound on λ_- using the seed set $\mathcal{S}_0$, and then apply a matrix concentration inequality to lower bound $\lambda_{\min}(M_{\tau+1})$.

Step 1: $\lambda_- > 0$. We establish a strictly positive lower bound on λ_- in the following sub-steps.

Seed set is always available in the sampling pool. By Assumption 2.4(iii), $\mathcal{S}_0 \subseteq X_t$ holds almost surely under $\mathcal{D}$. Recall the pure exploration safe candidate set

$$X^w := \Big\{ x \in X_t : \max_{\theta \in \Theta} \theta^\top B x \le c \Big\} \cup \mathcal{S}_0, \tag{23}$$

where $\Theta = \{\theta \in \mathbb{R}^d : \|\theta\|_2 \le S\}$. By Cauchy–Schwarz, $\theta^\top Bx \le \|\theta\|_2 \|Bx\|_2$, with equality when θ is parallel to Bx and $\|\theta\|_2 = S$. Hence $\max_{\theta\in\Theta} \theta^\top Bx = S\|Bx\|_2$, so the first set in (23) equals $\{x \in X_t : \|Bx\|_2 \le c/S\}$. By the explicit union with $\mathcal{S}_0$ in (23), we get $\mathcal{S}_0 \subseteq X^w$ almost surely.

Probability of sampling a pair entirely from $\mathcal{S}_0$. At each round, the sampling rule $\pi_{\exp}(X_t)$ in line 4 of Algorithm 1 draws an ordered pair (x, y) uniformly without replacement from X^w. Since $|X^w| \le |X_t \cup \mathcal{S}_0| \le n$ (using Assumption 2.4(iii) so $\mathcal{S}_0 \subseteq X_t$ and $|X_t| = n$, hence $X_t \cup \mathcal{S}_0 = X_t$ and $|X^w| \le n$), and $\mathcal{S}_0 \subseteq X^w$ a.s., the probability that both x and y fall inside $\mathcal{S}_0$ is at least

$$p_{\min} \;:=\; \frac{|\mathcal{S}_0|(|\mathcal{S}_0| - 1)}{n(n-1)} \;>\; 0, \tag{24}$$

where positivity follows from $|\mathcal{S}_0| \ge \max(2, d) \ge 2$ in Assumption 2.4.

PSD lower bound via total expectation. Let $\mathcal{E} := \{x \in \mathcal{S}_0 \text{ and } y \in \mathcal{S}_0\}$ be the event that both sampled arms belong to the seed set, and let $\mathcal{E}^c$ denote its complement. By the law of total expectation,

$$\mathbb{E}\big[(x-y)(x-y)^\top\big] = \mathbb{P}(\mathcal{E}) \cdot \mathbb{E}\big[(x-y)(x-y)^\top \mid \mathcal{E}\big] + \mathbb{P}(\mathcal{E}^c) \cdot \mathbb{E}\big[(x-y)(x-y)^\top \mid \mathcal{E}^c\big].$$

The matrix $(x-y)(x-y)^\top$ is a rank-one positive semidefinite (PSD) matrix for any realization of (x, y), so the conditional expectation on $\mathcal{E}^c$ is also PSD. Discarding this non-negative term and using $\mathbb{P}(\mathcal{E}) \ge p_{\min}$ from (24),

$$\mathbb{E}\big[(x-y)(x-y)^\top\big] \;\succeq\; p_{\min} \cdot \mathbb{E}\big[(x-y)(x-y)^\top \mid \mathcal{E}\big]. \tag{25}$$

Conditioned on $\mathcal{E}$, (x, y) is uniformly distributed over ordered pairs in $\mathcal{S}_0 \times \mathcal{S}_0$ with $x \ne y$. Hence

$$\mathbb{E}\big[(x-y)(x-y)^\top \mid \mathcal{E}\big] = \mathbb{E}_{x,y\sim \mathrm{Unif}(\mathcal{S}_0),\, x\ne y}\big[(x-y)(x-y)^\top\big].$$

Closed-form computation of the conditional expectation. Let $k := |\mathcal{S}_0|$, $\mu_{\mathcal{S}_0} := \frac{1}{k}\sum_{x\in\mathcal{S}_0} x$, and $\Sigma_{\mathcal{S}_0} := \frac{1}{k}\sum_{x\in\mathcal{S}_0} xx^\top$. Expanding the outer product,

$$\mathbb{E}\big[(x-y)(x-y)^\top \mid \mathcal{E}\big] = \underbrace{\mathbb{E}[xx^\top]}_{\text{(A)}} + \underbrace{\mathbb{E}[yy^\top]}_{\text{(B)}} - \underbrace{\mathbb{E}[xy^\top]}_{\text{(C)}} - \underbrace{\mathbb{E}[yx^\top]}_{\text{(D)}}. \tag{26}$$

We compute each term.

Terms (A) and (B). Under uniform sampling without replacement, the marginal distribution of x (resp. y) is uniform over $\mathcal{S}_0$ (the restriction $x \ne y$ does not change the marginal by symmetry). Hence

$$\mathbb{E}[xx^\top] = \mathbb{E}[yy^\top] = \Sigma_{\mathcal{S}_0}.$$

Terms (C) and (D). Since (x, y) is uniform over the $k(k-1)$ ordered pairs in $\mathcal{S}_0 \times \mathcal{S}_0$ with $x \ne y$,

$$\mathbb{E}[xy^\top] = \frac{1}{k(k-1)} \sum_{\substack{u,v\in\mathcal{S}_0 \\ u\ne v}} uv^\top = \frac{1}{k(k-1)} \left[\Big(\sum_{u\in\mathcal{S}_0} u\Big)\Big(\sum_{v\in\mathcal{S}_0} v\Big)^\top - \sum_{u\in\mathcal{S}_0} uu^\top \right].$$

Substituting $\sum_u u = k\mu_{\mathcal{S}_0}$ and $\sum_u uu^\top = k\Sigma_{\mathcal{S}_0}$,

$$\mathbb{E}[xy^\top] = \frac{k^2 \mu_{\mathcal{S}_0}\mu_{\mathcal{S}_0}^\top - k\Sigma_{\mathcal{S}_0}}{k(k-1)} = \frac{k\mu_{\mathcal{S}_0}\mu_{\mathcal{S}_0}^\top - \Sigma_{\mathcal{S}_0}}{k-1}.$$

By symmetry of the joint distribution, $\mathbb{E}[yx^\top] = \mathbb{E}[xy^\top]$ (both equal the same scalar quantity since $\mu_{\mathcal{S}_0}\mu_{\mathcal{S}_0}^\top$ and $\Sigma_{\mathcal{S}_0}$ are themselves symmetric matrices).

Combining. Substituting (A), (B), (C), (D) into (26),

$$\begin{aligned}\mathbb{E}\big[(x-y)(x-y)^\top \mid \mathcal{E}\big] &= 2\Sigma_{\mathcal{S}_0} - 2\cdot\frac{k\mu_{\mathcal{S}_0}\mu_{\mathcal{S}_0}^\top - \Sigma_{\mathcal{S}_0}}{k-1}\\ &= \frac{2(k-1)\Sigma_{\mathcal{S}_0} - 2k\mu_{\mathcal{S}_0}\mu_{\mathcal{S}_0}^\top + 2\Sigma_{\mathcal{S}_0}}{k-1}\\ &= \frac{2k\Sigma_{\mathcal{S}_0} - 2k\mu_{\mathcal{S}_0}\mu_{\mathcal{S}_0}^\top}{k-1}\\ &= \frac{2k}{k-1}\big(\Sigma_{\mathcal{S}_0} - \mu_{\mathcal{S}_0}\mu_{\mathcal{S}_0}^\top\big).\end{aligned}$$

Substituting $k = |\mathcal{S}_0|$,

$$\mathbb{E}\big[(x-y)(x-y)^\top \mid \mathcal{E}\big] = \frac{2|\mathcal{S}_0|}{|\mathcal{S}_0|-1}\big(\Sigma_{\mathcal{S}_0} - \mu_{\mathcal{S}_0}\mu_{\mathcal{S}_0}^\top\big). \tag{27}$$

Spectral lower bound and conclusion. By Assumption 2.4(ii), $\Sigma_{\mathcal{S}_0} \succeq \rho_0 I_d$. For the centered version, observe that $\mu_{\mathcal{S}_0}\mu_{\mathcal{S}_0}^\top$ is a rank-one PSD matrix. If the seed set $\mathcal{S}_0$ is chosen to be symmetric about the origin, i.e., $x \in \mathcal{S}_0 \iff -x \in \mathcal{S}_0$, then $\mu_{\mathcal{S}_0} = 0$, and

$$\Sigma_{\mathcal{S}_0} - \mu_{\mathcal{S}_0}\mu_{\mathcal{S}_0}^\top = \Sigma_{\mathcal{S}_0} \succeq \rho_0 I_d.$$

For general (non-symmetric) seed sets, as long as $\|\mu_{\mathcal{S}_0}\|_2^2 \le \rho_0/2$, we still obtain

$$\Sigma_{\mathcal{S}_0} - \mu_{\mathcal{S}_0}\mu_{\mathcal{S}_0}^\top \succeq \rho_0 I_d - \|\mu_{\mathcal{S}_0}\|_2^2 \cdot I_d \succeq \tfrac{\rho_0}{2} I_d.$$

In either case, $\lambda_{\min}(\Sigma_{\mathcal{S}_0} - \mu_{\mathcal{S}_0}\mu_{\mathcal{S}_0}^\top) \ge \rho_0/2$. Combining with (25) and (27),

$$\begin{aligned}\lambda_- = \lambda_{\min}\big(\mathbb{E}[(x-y)(x-y)^\top]\big) &\ge p_{\min}\cdot\lambda_{\min}\Big(\tfrac{2|\mathcal{S}_0|}{|\mathcal{S}_0|-1}(\Sigma_{\mathcal{S}_0} - \mu_{\mathcal{S}_0}\mu_{\mathcal{S}_0}^\top)\Big)\\ &\ge p_{\min}\cdot\frac{2|\mathcal{S}_0|}{|\mathcal{S}_0|-1}\cdot\frac{\rho_0}{2}\\ &= \frac{p_{\min}\,|\mathcal{S}_0|\,\rho_0}{|\mathcal{S}_0|-1}.\end{aligned}$$

Therefore,

$$\lambda_- \;\ge\; \frac{p_{\min}\,|\mathcal{S}_0|\,\rho_0}{|\mathcal{S}_0|-1} \;>\; 0, \tag{28}$$

which establishes the desired strictly positive lower bound on λ_-.

Step 2: Lower bounding $\lambda_{\min}(M_{\tau+1})$. By the stochastic contextual setting described in Section 2, $\{X_t\}_{t\ge1}$ are i.i.d. samples from $\mathcal{D}$. Combined with the memoryless uniform sampling in line 4 of Algorithm 1, the difference vectors $\{z_t\}_{t=1}^\tau$ are i.i.d. across t. To control the minimum eigenvalue of the Gram matrix, we apply the matrix Chernoff inequality.

Theorem B.2 (Matrix Chernoff Inequality)**.** *Consider a finite sequence $\{X_k\}$ of independent, symmetric random matrices in $\mathbb{R}^{d\times d}$ such that $\lambda_{\min}(X_k) \ge 0$ and $\lambda_{\max}(X_k) \le L$ almost surely. Let $Y = \sum_k X_k$ and $\mu_{\min} := \lambda_{\min}(\mathbb{E}[Y])$. Then for any $\epsilon \in (0,1)$,*

$$\Pr\big(\lambda_{\min}(Y) \le \epsilon\mu_{\min}\big) \le d\cdot\exp\Big(-\frac{(1-\epsilon)^2\mu_{\min}}{2L}\Big).$$

Let $Z_t := z_t z_t^\top$ for $t \in [\tau]$. Each Z_t is symmetric and positive semidefinite, and satisfies

$$\lambda_{\max}(Z_t) \;=\; \|z_t\|_2^2 \;\le\; (\|x_{t,i_t}\|_2 + \|x_{t,j_t}\|_2)^2 \;\le\; 4.$$

Moreover, $M_{\tau+1} = \sum_{t=1}^\tau Z_t$, and

$$\mu_{\min} \;:=\; \lambda_{\min}\Big(\sum_{t=1}^{\tau}\mathbb{E}[Z_t]\Big) \;=\; \lambda_{\min}(\tau\Sigma) \;=\; \tau\lambda_-.$$

Applying Theorem B.2 with $L = 4$ yields, for any $\epsilon \in (0, 1)$,

$$\Pr\Big(\lambda_{\min}(M_{\tau+1}) \le \epsilon\lambda_-\tau\Big) \le d \cdot \exp\Big(-\frac{(1-\epsilon)^2\lambda_-\tau}{8}\Big).$$

Setting $\epsilon = \frac{1}{2}$ and choosing

$$\tau \;\ge\; \frac{32}{\lambda_-}\log\Big(\frac{d}{\delta}\Big)$$

ensures that the right-hand side is at most δ. Hence, with probability at least $1 - \delta$,

$$\lambda_{\min}(M_{\tau+1}) \;\ge\; \frac{\lambda_-\tau}{2},$$

which completes the proof. □

C Bounding Term I

To control Term I, which corresponds to the standard exploration–exploitation error under safety constraints, we introduce several high-probability concentration events associated with the Safe-COLSTIM algorithm. These events ensure that the estimated utilities, their perturbed counterparts, and the true utilities remain well aligned throughout the learning process.

(1) Maximum-likelihood estimation event A_{MLE}. There exist constants $c_1 > 0$ such that

$$A_{\mathrm{MLE}} := \Big\{\forall\{i,j\} \subset [n],\ t > \tau : \big|\hat{u}_{t,i}(X_t) - \hat{u}_{t,j}(X_t) - \big(u_i^*(X_t) - u_j^*(X_t)\big)\big| \le c_1 \left\|z_{t,i,j}\right\|_{M_t^{-1}}\Big\}.$$

where $\hat{u}_{t,i}(X_t) = x_{t,i}^\top\hat{\theta}_t$ and $u_i^*(X_t) = x_{t,i}^\top\theta^*$. This event guarantees a uniform concentration of the estimated pairwise utility gaps.

(2) Perturbation concentration event $A_{\mathrm{conc},t}$. For each $t > \tau$, define

$$A_{\mathrm{conc},t} := \Big\{\forall\{i,j\} \subset [n] : \big|\big(\tilde{u}_{t,i}(X_t) - \tilde{u}_{t,j}(X_t)\big) - \big(\hat{u}_{t,i}(X_t) - \hat{u}_{t,j}(X_t)\big)\big| \le c_2 \left\|z_{t,i,j}\right\|_{M_t^{-1}}\Big\}. \tag{29}$$

where $\tilde{u}_{t,i}(X_t) = \hat{u}_{t,i}(X_t) + \epsilon_{t,i}\|x_{t,i}\|_{M_t^{-1}}$. This event controls the magnitude of the randomized perturbations used for exploration.

(3) Initialization event A_{init}. We further define

$$A_{\mathrm{init}} := \Big\{\forall t > \tau : \|\hat{\theta}_t - \theta^*\|_2 \le 1 \text{ and } M_t \succeq I_d\Big\},$$

which ensures estimator stability and invertibility of the information matrix after the pure exploration phase.

(4) Regret contribution of Term I. Let (i_t, j_t) be the queried arm pair at round t. The instantaneous average regret under Term I is

$$\Delta_{(i_t,j_t)} := \frac{\langle x_{t,i_S^*}, \theta^*\rangle - \langle x_{t,i_t}, \theta^*\rangle + \langle x_{t,i_S^*}, \theta^*\rangle - \langle x_{t,j_t}, \theta^*\rangle}{2}.$$

Theorem C.1 (Term I regret bound)**.** *Let $(S_t)_{t\in[T]} = (i_t, j_t)$ be the sequence selected by Safe-COLSTIM. Assume that*

$$\mathbb{P}(A_{\mathrm{MLE}}) \ge 1 - p_1, \qquad \mathbb{P}(A_{\mathrm{init}}) \ge 1 - p_2,$$

and for all $t > \tau$,

$$\mathbb{P}(A_{\mathrm{conc},t} \mid \mathcal{H}_{t-1}) \ge 1 - p_{3,t}.$$

Then, for any $c_3 \ge \sqrt{2d}$,

$$\mathbb{E}[R_T^a(\text{Term I})] \le\; \Delta_{\max}\tau + \Delta_{\max}(p_1 + p_2)T + \Delta_{\max}\sum_{t=\tau+1}^{T} p_{3,t} + \frac{c_3}{2}(3c_1 + c_2)\sqrt{T\log\Big(\frac{2T}{d}\Big)},$$

where $\Delta_{\max} = 2\sqrt{2}$.

Moreover, if

$$\sum_{t=\tau+1}^{T} \lambda_{\min}^{-1/2}(M_t) \le b\sqrt{T},$$

then the bound holds for any $c_3 \ge \sqrt{2}\, b$.

To prove Theorem C.1, we first establish two auxiliary lemmas.

Lemma C.2. *Let* $z_1, z_2, \ldots, z_t \in \mathbb{R}^d$ *satisfy* $\max_{s\in[t]} \|z_s\|^2 \le 2$, *and define*

$$M_{t+1} = \sum_{s=1}^{t} z_s z_s^\top.$$

Assume further that $t > \tau$ *and* $M_{\tau+1} \succeq I_d$. *Then,*

$$\sum_{s=\tau+1}^{t} \left(\sqrt{2} \wedge \|z_s\|_{M_s^{-1}}\right) \le \sqrt{2dt \log\left(\frac{2t}{d}\right)}.$$

Proof. We first note that the inequality $x \le 2\log(1+x)$ holds for all $x \in [0,2]$, which applies here due to the assumption $\max_{s\in[t]} \|z_s\|^2 \le 2$. Therefore,

$$\begin{aligned}
\sum_{s=\tau+1}^{t} \left(\sqrt{2} \wedge \|z_s\|_{M_s^{-1}}\right)^2 &\le \sum_{s=\tau+1}^{t} \|z_s\|^2_{M_s^{-1}} \\
&\le \sum_{s=\tau+1}^{t} 2\log\left(1 + \|z_s\|^2_{M_s^{-1}}\right) \\
&= 2\log\left(\prod_{s=\tau+1}^{t} \left(1 + \|z_s\|^2_{M_s^{-1}}\right)\right). \qquad (30)
\end{aligned}$$

Let $\det(A)$ denote the determinant of a square matrix A. By the matrix determinant lemma, we have

$$\det(M_{t+1}) = \det\left(M_t + z_t z_t^\top\right) = \det(M_t)\, \det\left(I + M_t^{-1} z_t z_t^\top\right) = \det(M_t)\left(1 + z_t^\top M_t^{-1} z_t\right).$$

where the last equality follows from the rank-one determinant identity $\det(I + uv^\top) = 1 + v^\top u$. Iterating this recursion yields

$$\begin{aligned}
\det(M_{t+1}) &= \det(M_{\tau+1}) \prod_{s=\tau+1}^{t} \left(1 + z_s^\top M_s^{-1} z_s\right) \\
&= \det(M_{\tau+1}) \prod_{s=\tau+1}^{t} \left(1 + \|z_s\|^2_{M_s^{-1}}\right) \\
&\ge \prod_{s=\tau+1}^{t} \left(1 + \|z_s\|^2_{M_s^{-1}}\right), \qquad (31)
\end{aligned}$$

where the last inequality uses $M_{\tau+1} \succeq I_d$.

Next, applying the determinant–trace inequality for positive definite matrices,

$$\det(A)^{1/d} \le \frac{\mathrm{tr}(A)}{d},$$

we obtain

$$\det(M_{t+1})^{1/d} \le \frac{\mathrm{tr}(M_{t+1})}{d} = \frac{\mathrm{tr}\left(\sum_{s=1}^{t} z_s z_s^\top\right)}{d} = \frac{\sum_{s=1}^{t} \|z_s\|^2}{d} \le \frac{2t}{d}, \qquad (32)$$

which implies $\det(M_{t+1}) \le (2t/d)^d$.

Combining (30), (31), and (32), we obtain

$$\sum_{s=\tau+1}^{t} \left(\sqrt{2} \wedge \|z_s\|_{M_s^{-1}}\right)^2 \le 2\log\big(\det(M_{t+1})\big) \le 2d\log\left(\frac{2t}{d}\right).$$

Finally, applying the Cauchy–Schwarz inequality yields

$$\sum_{s=\tau+1}^{t} \left(\sqrt{2} \wedge \|z_s\|_{M_s^{-1}}\right) \le \sqrt{(t-\tau)\sum_{s=\tau+1}^{t} \left(\sqrt{2} \wedge \|z_s\|_{M_s^{-1}}\right)^2} \le \sqrt{2dt\log\left(\frac{2t}{d}\right)}.$$

which completes the proof. □

Lemma C.3 (Single-round regret control)**.** *Under the assumptions of Theorem C.1, for any $t > \tau$ and any history $\mathcal{H}_{t-1}$,*

$$\mathbb{E}[\Delta_{i_t,j_t}\mathbf{1}_{A_{\mathrm{MLE}}\cap A_{\mathrm{init}}} \mid \mathcal{H}_{t-1}] \le \Delta_{\max} p_{3,t} + \frac{3c_1+c_2}{2}\mathbb{E}\Big[\|z_{t,i_t,j_t}\|_{M_t^{-1}}\mathbf{1}_{A_{\mathrm{init}}} \mid \mathcal{H}_{t-1}\Big].$$

Proof. Fix any t and history $\mathcal{H}_{t-1}$ such that A_{MLE} and A_{init} hold. We henceforth omit explicit conditioning on $\mathcal{H}_{t-1}$.

We first decompose the expectation:

$$\begin{aligned}
\mathbb{E}[\Delta_{i_t,j_t}\mathbf{1}_{A_{\mathrm{MLE}}\cap A_{\mathrm{init}}}] &= \mathbb{E}\big[\Delta_{i_t,j_t}\mathbf{1}_{A_{\mathrm{MLE}}\cap A_{\mathrm{init}}}\mathbf{1}_{A_{\mathrm{conc},t}}\big] + \mathbb{E}\Big[\Delta_{i_t,j_t}\mathbf{1}_{A_{\mathrm{MLE}}\cap A_{\mathrm{init}}}\mathbf{1}_{A_{\mathrm{conc},t}^{\mathsf{C}}}\Big] \\
&\le \mathbb{E}\big[\Delta_{i_t,j_t}\mathbf{1}_{A_{\mathrm{MLE}}\cap A_{\mathrm{init}}}\mathbf{1}_{A_{\mathrm{conc},t}}\big] + \Delta_{\max}\,\mathbb{P}\big(A_{\mathrm{conc},t}^{\mathsf{C}}\big) \\
&\le \mathbb{E}\big[\Delta_{i_t,j_t}\mathbf{1}_{A_{\mathrm{MLE}}\cap A_{\mathrm{init}}}\mathbf{1}_{A_{\mathrm{conc},t}}\big] + \Delta_{\max} p_{3,t}.
\end{aligned}$$

Under event A_{MLE}, we have

$$\langle z_{t,i_S^*,i_t}, \theta^* - \hat{\theta}_t\rangle \le c_1\|z_{t,i_S^*,i_t}\|_{M_t^{-1}}, \langle z_{t,i_S^*,j_t}, \theta^* - \hat{\theta}_t\rangle \le c_1\|z_{t,i_S^*,j_t}\|_{M_t^{-1}}. \tag{33}$$

Noting that

$$z_{t,i_S^*,j_t} = z_{t,i_S^*,i_t} + z_{t,i_t,j_t}, \tag{34}$$

and using the definition $j_t = \arg\max_{i\in\mathcal{A}_t}\langle z_{t,i_t,i}, \hat{\theta}_t\rangle + c_1\|z_{t,i_t,i}\|_{M_t^{-1}}$, we obtain for all $i \in \mathcal{A}_t$,

$$\langle z_{t,i,i_t}, \hat{\theta}_t\rangle + c_1\|z_{t,i,i_t}\|_{M_t^{-1}} \le \langle z_{t,j_t,i_t}, \hat{\theta}_t\rangle + c_1\|z_{t,i_t,j_t}\|_{M_t^{-1}}. \tag{35}$$

Similarly, by the definition $i_t = \arg\max_{i\in\mathcal{A}_t}\langle x_{t,i}, \hat{\theta}_t\rangle + \epsilon_{t,i}\|x_{t,i}\|_{M_t^{-1}}$, we have for all $i \in \mathcal{A}_t$,

$$\langle z_{t,i,i_t}, \hat{\theta}_t\rangle + \epsilon_{t,i}\|x_{t,i}\|_{M_t^{-1}} - \epsilon_{t,i_t}\|x_{t,i_t}\|_{M_t^{-1}} \le 0. \tag{36}$$

With these preparations, we bound

$$\begin{aligned}
2\Delta_{i_t,j_t} &= \langle z_{t,i_S^*,i_t}, \theta^*\rangle + \langle z_{t,i_S^*,j_t}, \theta^*\rangle \\
&= \langle z_{t,i_S^*,i_t}, \theta^* - \hat{\theta}_t\rangle + \langle z_{t,i_S^*,i_t}, \hat{\theta}_t\rangle + \langle z_{t,i_S^*,j_t}, \theta^* - \hat{\theta}_t\rangle + \langle z_{t,i_S^*,j_t}, \hat{\theta}_t\rangle \\
&\overset{(33)}{\le} c_1\Big(\|z_{t,i_S^*,i_t}\|_{M_t^{-1}} + \|z_{t,i_S^*,j_t}\|_{M_t^{-1}}\Big) + \langle z_{t,i_S^*,i_t}, \hat{\theta}_t\rangle + \langle z_{t,i_S^*,j_t}, \hat{\theta}_t\rangle \\
&\overset{(34)}{=} c_1\Big(\|z_{t,i_S^*,i_t}\|_{M_t^{-1}} + \|z_{t,i_S^*,i_t} + z_{t,i_t,j_t}\|_{M_t^{-1}}\Big) + 2\langle z_{t,i_S^*,i_t}, \hat{\theta}_t\rangle + \langle z_{t,i_t,j_t}, \hat{\theta}_t\rangle \\
&\le c_1\Big(2\|z_{t,i_S^*,i_t}\|_{M_t^{-1}} + \|z_{t,i_t,j_t}\|_{M_t^{-1}}\Big) + 2\langle z_{t,i_S^*,i_t}, \hat{\theta}_t\rangle + \langle z_{t,i_t,j_t}, \hat{\theta}_t\rangle \\
&\overset{(35)}{\le} 3c_1\|z_{t,i_t,j_t}\|_{M_t^{-1}} + 2\langle z_{t,j_t,i_t}, \hat{\theta}_t\rangle + \langle z_{t,i_t,j_t}, \hat{\theta}_t\rangle \\
&= 3c_1\|z_{t,i_t,j_t}\|_{M_t^{-1}} + \langle z_{t,j_t,i_t}, \hat{\theta}_t\rangle \\
&\overset{(36)}{\le} 3c_1\|z_{t,i_t,j_t}\|_{M_t^{-1}} + \epsilon_{t,i_t}\|x_{t,i_t}\|_{M_t^{-1}} - \epsilon_{t,j_t}\|x_{t,j_t}\|_{M_t^{-1}} \\
&\le (3c_1+c_2)\|z_{t,i_t,j_t}\|_{M_t^{-1}},
\end{aligned}$$

where the last inequality follows from $A_{\mathrm{conc},t}$.

Therefore,

$$\begin{aligned}
&\mathbb{E}[\Delta_{i_t,j_t}\mathbf{1}_{A_{\mathrm{MLE}}\cap A_{\mathrm{init}}} \mid \mathcal{H}_{t-1}] \\
&\le \mathbb{E}\big[\Delta_{i_t,j_t}\mathbf{1}_{A_{\mathrm{MLE}}\cap A_{\mathrm{init}}}\mathbf{1}_{A_{\mathrm{conc},t}} \mid \mathcal{H}_{t-1}\big] + \Delta_{\max}p_{3,t} \\
&\le \frac{3c_1+c_2}{2}\mathbb{E}\Big[\|z_{t,i_t,j_t}\|_{M_t^{-1}}\mathbf{1}_{A_{\mathrm{init}}} \mid \mathcal{H}_{t-1}\Big] + \Delta_{\max}p_{3,t} \\
&= \Delta_{\max}p_{3,t} + \frac{3c_1+c_2}{2}\mathbb{E}\Big[\|z_{t,i_t,j_t}\|_{M_t^{-1}}\mathbf{1}_{A_{\mathrm{init}}} \mid \mathcal{H}_{t-1}\Big],
\end{aligned}$$

which completes the proof. □

Proof of Theorem C.1. Let $\Delta_{\max} = 2\sqrt{2}$. Then,

$$2\Delta_{(i_t,j_t)} = \langle x_{t,i_S^*},\theta^*\rangle - \langle x_{t,i_t},\theta^*\rangle + \langle x_{t,i_S^*},\theta^*\rangle - \langle x_{t,j_t},\theta^*\rangle \le \Delta_{\max}.$$

The above inequality follows from the assumptions $\max\{\|x_{t,i}\|^2, \|x_{t,j}\|^2\} \le 1$ for all i, j, which implies $\|z_{t,i,j}\|^2 \le 2$ for any $i, j \in \mathcal{A}_t$, together with $\|\theta^*\| \le 1$ and $\|\theta - \theta^*\| \le 1$. Applying the triangle inequality and the Cauchy–Schwarz inequality yields

$$\begin{aligned}
&\langle x_{t,i_S^*},\theta^*\rangle - \langle x_{t,i_t},\theta^*\rangle + \langle x_{t,i_S^*},\theta^*\rangle - \langle x_{t,j_t},\theta^*\rangle \\
&= \langle z_{t,i_S^*,i_t},\theta^*\rangle + \langle z_{t,i_S^*,j_t},\theta^*\rangle \le \big|\langle z_{t,i_S^*,i_t},\theta^*\rangle\big| + \big|\langle z_{t,i_S^*,j_t},\theta^*\rangle\big| \\
&\le \|z_{t,i_S^*,i_t}\|\,\|\theta^*\| + \|z_{t,i_S^*,j_t}\|\,\|\theta^*\| \le \sqrt{2}+\sqrt{2} = 2\sqrt{2} = \Delta_{\max}.
\end{aligned}$$

Consequently, the expected cumulative average regret under Term I satisfies

$$\begin{aligned}
\mathbb{E}[R_T^a(\text{Term I})] &= \sum_{t=1}^{T}\mathbb{E}[\Delta_{i_t,j_t}] \\
&\le \Delta_{\max}\tau + \sum_{t=\tau+1}^{T}\mathbb{E}[\Delta_{i_t,j_t}] \\
&\le \Delta_{\max}\tau + \Delta_{\max}T\Big(\mathbb{P}(A_{\mathrm{init}}^c) + \mathbb{P}(A_{\mathrm{MLE}}^c)\Big) + \sum_{t=\tau+1}^{T}\mathbb{E}[\Delta_{i_t,j_t}\mathbf{1}_{A_{\mathrm{MLE}}\cap A_{\mathrm{init}}}],
\end{aligned}$$

where the second inequality follows from the law of total expectation:

$$\mathbb{E}[\Delta_{i_t,j_t}] = \mathbb{E}[\Delta_{i_t,j_t} \mid (A_{\mathrm{MLE}}\cap A_{\mathrm{init}})^c]\,\mathbb{P}((A_{\mathrm{MLE}}\cap A_{\mathrm{init}})^c) + \mathbb{E}[\Delta_{i_t,j_t} \mid A_{\mathrm{MLE}}\cap A_{\mathrm{init}}]\,\mathbb{P}(A_{\mathrm{MLE}}\cap A_{\mathrm{init}}).$$

Finally, by the tower property of conditional expectation,

$$\sum_{t=\tau+1}^{T}\mathbb{E}[\Delta_{i_t,j_t}\mathbf{1}_{A_{\mathrm{MLE}}\cap A_{\mathrm{init}}}] = \sum_{t=\tau+1}^{T}\mathbb{E}[\mathbb{E}[\Delta_{i_t,j_t}\mathbf{1}_{A_{\mathrm{MLE}}\cap A_{\mathrm{init}}} \mid \mathcal{H}_{t-1}]].$$

Next, applying Lemma C.3, we obtain

$$\begin{aligned}
\mathbb{E}[R_T^a(\text{Term I})] &\le \Delta_{\max}\tau + \Delta_{\max}T\Big(\mathbb{P}(A_{\mathrm{init}}^c) + \mathbb{P}(A_{\mathrm{MLE}}^c)\Big) + \sum_{t=\tau+1}^{T}\mathbb{E}\big[\mathbb{E}\big[\Delta_{(i_t,j_t)}\mathbf{1}_{A_{\mathrm{MLE}}\cap A_{\mathrm{init}}} \mid \mathcal{H}_{t-1}\big]\big] \\
&\le \Delta_{\max}\tau + \Delta_{\max}T\Big(\mathbb{P}(A_{\mathrm{init}}^c) + \mathbb{P}(A_{\mathrm{MLE}}^c)\Big) + \Delta_{\max}\sum_{t=\tau+1}^{T}p_{3,t} \\
&\quad + \frac{3c_1+c_2}{2}\sum_{t=\tau+1}^{T}\mathbb{E}\Big[\mathbb{E}\Big[\|z_{t,i_t,j_t}\|_{M_t^{-1}}\mathbf{1}_{A_{\mathrm{init}}} \mid \mathcal{H}_{t-1}\Big]\Big].
\end{aligned}$$

Under the initialization event

$$A_{\text{init}} = \Big\{\forall t > \tau : \|\hat{\theta}_t - \theta^*\| \le 1 \text{ and } M_t \succeq I_d\Big\},$$

we have $M_t^{-1} \preceq I_d$. Therefore, for any $i, j \in \mathcal{A}_t$ and any $t > \tau$,

$$\|z_{t,i,j}\|_{M_t^{-1}} = \sqrt{z_{t,i,j}^\top M_t^{-1} z_{t,i,j}} \le \sqrt{z_{t,i,j}^\top z_{t,i,j}} = \|z_{t,i,j}\| \le \sqrt{2}.$$

We can now use Lemma C.2 to obtain

$$\mathbb{E}[R_T^a(\text{Term I})] \le \Delta_{\max}\tau + \Delta_{\max} T\Big(\mathbb{P}(A_{\text{init}}^c) + \mathbb{P}(A_{\text{MLE}}^c)\Big) + \Delta_{\max}\sum_{t=1}^{T} p_{3,t} + \frac{c_3}{2}(3c_1 + c_2)\sqrt{T\log\left(\frac{2T}{d}\right)},$$

for any $c_3 \ge \sqrt{2d}$.

By the assumptions of the theorem, $\mathbb{P}(A_{\text{init}}^c) \le p_2$ and $\mathbb{P}(A_{\text{MLE}}^c) \le p_1$. Substituting these bounds yields

$$\mathbb{E}[R_T^a(\text{Term I})] \le \Delta_{\max}\tau + \Delta_{\max}(p_1 + p_2)T + \Delta_{\max}\sum_{t=\tau+1}^{T} p_{3,t} + \frac{c_3}{2}(3c_1 + c_2)\sqrt{T\log\left(\frac{2T}{d}\right)}.$$

For the second statement, note that by the Cauchy–Schwarz inequality

$$\begin{aligned}
\sum_{t=\tau+1}^{T}\Big(\sqrt{2} \wedge \|z_{t,i_t,j_t}\|_{M_t^{-1}}\Big) &\le \sum_{t=\tau+1}^{T}\sqrt{z_{t,i_t,j_t}^\top M_t^{-1} z_{t,i_t,j_t}} \\
&= \sum_{t=\tau+1}^{T}\|M_t^{-1/2} z_{t,i_t,j_t}\| \\
&\le \sum_{t=\tau+1}^{T}\|z_{t,i_t,j_t}\|\,\|M_t^{-1/2}\| \\
&\le \sqrt{2}\sum_{t=\tau+1}^{T}\lambda_{\min}^{-1/2}(M_t).
\end{aligned}$$

The last inequality follows from the assumption $\|z_{t,i,j}\|^2 \le 2$ for all $i, j \in \mathcal{A}$, and from the spectral norm identity

$$\|M\| = \sqrt{\lambda_{\max}(M^\top M)}, \qquad \|M_t^{-1/2}\| = \frac{1}{\sqrt{\lambda_{\min}(M_t)}}.$$

Therefore, under the assumption that

$$\sum_{t=\tau+1}^{T}\lambda_{\min}^{-1/2}(M_t) \le b\sqrt{T},$$

we obtain

$$\sum_{t=\tau+1}^{T}\Big(\sqrt{2} \wedge \|z_{t,i_t,j_t}\|_{M_t^{-1}}\Big) \le b\sqrt{2T}.$$

Combining the above bounds, we obtain

$$\begin{aligned}
\mathbb{E}[R_T^a(\text{Term I})] &\le \Delta_{\max}\tau + \Delta_{\max} T\Big(\mathbb{P}(A_{\text{init}}^c) + \mathbb{P}(A_{\text{MLE}}^c)\Big) + \Delta_{\max}\sum_{t=1}^{T} p_{3,t} \\
&\quad + \frac{3c_1 + c_2}{2}\sum_{t=\tau+1}^{T}\mathbb{E}\Big[\mathbb{E}\big[\|z_{t,i_t,j_t}\|_{M_t^{-1}}\mathbf{1}_{A_{\text{init}}} \mid \mathcal{H}_{t-1}\big]\Big] \\
&\le \Delta_{\max}\tau + \Delta_{\max} T\Big(\mathbb{P}(A_{\text{init}}^c) + \mathbb{P}(A_{\text{MLE}}^c)\Big) + \Delta_{\max}\sum_{t=1}^{T} p_{3,t} + \frac{1}{2}(3c_1 + c_2)\, b\sqrt{2T}.
\end{aligned}$$

Rewriting the last term, we have

$$\frac{1}{2}(3c_1+c_2)\,b\sqrt{2T} = \frac{\sqrt{2}b}{2}(3c_1+c_2)\sqrt{T} = \frac{\sqrt{2}b}{2}(3c_1+c_2)\sqrt{T\log(2T/d)}\cdot\frac{\sqrt{T}}{\sqrt{T\log(2T/d)}}.$$

Letting $c_3 := \sqrt{2}\,b$, we obtain

$$\mathbb{E}[R_T^a(\text{Term I})] \le \Delta_{\max}\tau + \Delta_{\max}T\Big(\mathbb{P}(A_{\text{init}}^c) + \mathbb{P}(A_{\text{MLE}}^c)\Big) + \Delta_{\max}\sum_{t=\tau+1}^{T} p_{3,t} + \frac{c_3}{2}(3c_1+c_2)\sqrt{T\log\left(\frac{2T}{d}\right)}.$$

This establishes the second statement of Theorem C.1. □

D Bounding Term II

In this section, we upper bound Term II in the regret of Safe-COLSTIM via the following key quantity, referred to as the *safety gap* Δ.

Bounding Term II is nontrivial since the estimated safe set X_t^s may differ from the true safe set X_0^s, and the optimal arm x_{t,i^*} need not lie in X_t^s. Our analysis controls this discrepancy through a notion of distance between X_t^s and X_0^s along the direction of x_{t,i^*}. The argument proceeds by introducing a conservative approximation of X_t^s, constructing a feasible proxy aligned with x_{t,i^*}, and bounding their separation.

A shrunken safe set. Consider the enlarged confidence region centered at the true parameter θ^*:

$$\widetilde{C}_t := \big\{w \in \mathbb{R}^d : \|w-\theta^*\|_{M_t} \le 2\beta_t\big\} \supseteq C_t := \big\{w \in \mathbb{R}^d : \|w-\hat{\theta}_t\|_{M_t} \le \beta_t\big\}.$$

The inclusion holds since $\theta^* \in C_t$ with high probability and, for any $w \in C_t$,

$$\|w-\theta^*\|_{M_t} \le \|w-\hat{\theta}_t\|_{M_t} + \|\hat{\theta}_t-\theta^*\|_{M_t} \le 2\beta_t.$$

This enlarged confidence region induces the following shrunken safe decision set:

$$\widetilde{X_t^s} := \big\{x \in X_0^s : w^\top Bx \le c,\ \forall w \in \widetilde{C}_t\big\} = \big\{x \in X_0^s : \theta^{*\top}Bx + 2\beta_t\|Bx\|_{M_t^{-1}} \le c\big\},$$

where the last equality follows by writing $w = \theta^* + \delta$ with $\|\delta\|_{M_t} \le 2\beta_t$ and applying the Cauchy–Schwarz inequality. By Assumption 2.2, both $\widetilde{X_t^s}$ and X_t^s have nonempty interiors.

A feasible proxy aligned with x_{t,i^*}. We define a point in $\widetilde{X_t^s}$ that lies in the direction of x_{t,i^*} and is closest to it:

$$v_t := \alpha_t x_{t,i^*}, \qquad \alpha_t := \max\big\{\alpha \in [0,1] : \alpha x_{t,i^*} \in \widetilde{X_t^s}\big\}.$$

Since $0, x_{t,i^*} \in X_0^s$ and X_0^s is convex, it follows that

$$\alpha_t = \max\Big\{\alpha \in [0,1] : \alpha\big(\theta^{*\top}Bx_{t,i^*} + 2\beta_t\|Bx_{t,i^*}\|_{M_t^{-1}}\big) \le c\Big\}.$$

Equivalently,

$$\alpha_t = \begin{cases} 1, & \text{if } \theta^{*\top}Bx_{t,i^*} + 2\beta_t\|Bx_{t,i^*}\|_{M_t^{-1}} \le c, \\ \dfrac{c}{\theta^{*\top}Bx_{t,i^*} + 2\beta_t\|Bx_{t,i^*}\|_{M_t^{-1}}}, & \text{otherwise.} \end{cases}$$

Bounding Term II via the safety scaling factor α_t. We now upper bound Term II in the regret of Safe-COLSTIM through the scaling factor α_t defined above. By construction, $\widetilde{X}_t^s \subseteq X_t^s$, and hence

$v_t = \alpha_t x_{t,i^*} \in X_t^s$. Using this fact together with the optimality of x_{t,i^*} over the true safe set X_0^s, we obtain

$$\begin{aligned}\text{Term II} &:= \langle x_{t,i^*}, \theta^* \rangle - \langle x_{t,i_S^*}, \theta^* \rangle \le \langle x_{t,i^*}, \theta^* \rangle - \langle v_t, \theta^* \rangle \\ &= (1-\alpha_t)\langle x_{t,i^*}, \theta^* \rangle \le |1-\alpha_t|\,|\langle x_{t,i^*}, \theta^* \rangle| \le 1-\alpha_t,\end{aligned}$$

where the last inequality follows from Assumption 2.1, which ensures $\langle x_{t,i^*}, \theta^* \rangle \in [-1, 1]$. Therefore, controlling Term II reduces to bounding $1 - \alpha_t$.

To further characterize the behavior of Term II, we distinguish two regimes depending on whether the safety gap Δ is positive or vanishes.

D.1 Vanishing safety-induced regret under a positive safety gap

When the safety gap $\Delta > 0$ is known, the safety-induced component of the regret eventually disappears once the pure exploration phase is sufficiently long. The following result formalizes this observation. Although the conclusion follows directly from Lemma 4.1 stated in the main text, we provide a self-contained proof to make explicit how the condition $\alpha_t = 1$ is ensured through the spectral properties of the information matrix.

Lemma D.1 (Positive safety gap implies vanishing Term II)**.** *Let Assumptions 2.1, 2.2, and 2.4 hold. Fix any $\delta \in (0, 0.5)$ and assume a positive safety gap $\Delta > 0$. Initialize Safe-COLSTIM such that*

$$\tau \ge T_\Delta := \Big(\frac{8\|B\|^2\beta_T^2}{\lambda_-\Delta^2}\Big) \ \vee \ t_\delta,$$

where t_δ is defined as in Lemma B.1. Then, with probability at least $1 - 2\delta$, for all $t = \tau + 1, \dots, T$,

$$\text{Term II} = \langle x_{t,i^*}, \theta^* \rangle - \langle x_{t,i_S^*}, \theta^* \rangle \le 0.$$

Consequently,

$$\sum_{t=\tau+1}^{T} \Big(\langle x_{t,i^*}, \theta^* \rangle - \langle x_{t,i_S^*}, \theta^* \rangle\Big) \le 0.$$

Proof. Recall that for any $\tau < t \le T$, with probability at least $1 - \delta$, we have

$$\text{Term II} = 1 - \alpha_t.$$

Thus, it suffices to show that for any $\tau < t \le T$, with probability at least $1 - \delta$, it holds that $\alpha_t = 1$. As established earlier, the condition $\alpha_t = 1$ is equivalent to

$$\theta^{*\top} B x_{t,i^*} + 2\beta_t \|B x_{t,i^*}\|_{M_t^{-1}} \le c.$$

Using the definition of the safety gap $\Delta := c - \theta^{*\top} B x_{t,i^*}$, this condition is equivalent to

$$\beta_t \|B x_{t,i^*}\|_{M_t^{-1}} \le \frac{\Delta}{2}.$$

Applying the Rayleigh quotient bound yields

$$\|B x_{t,i^*}\|_{M_t^{-1}} = \sqrt{(B x_{t,i^*})^\top M_t^{-1} (B x_{t,i^*})} \le \frac{\|B x_{t,i^*}\|_2}{\sqrt{\lambda_{\min}(M_t)}}.$$

Therefore, for any $\tau < t \le T$,

$$\beta_t \|B x_{t,i^*}\|_{M_t^{-1}} \le \frac{\beta_t \|B x_{t,i^*}\|_2}{\sqrt{\lambda_{\min}(M_t)}} \le \frac{\beta_T \|B x_{t,i^*}\|_2}{\sqrt{\lambda_{\min}(M_{\tau+1})}} \le \frac{\beta_T \|B\|}{\sqrt{\lambda_{\min}(M_{\tau+1})}}, \tag{37}$$

where we used $\beta_t \le \beta_T$, $\lambda_{\min}(M_t) \ge \lambda_{\min}(M_{\tau+1})$, and $\|B x_{t,i^*}\|_2 \le \|B\|$.

Since $t_\delta \le \tau$, Lemma B.1 implies that with probability at least $1 - \delta$,

$$\lambda_{\min}(M_{\tau+1}) \ge \frac{\lambda_- \tau}{2}.$$

Combining this with (37), we obtain, for all $t = \tau + 1, \dots, T$,

$$\beta_t \|B x_{t,i^*}\|_{M_t^{-1}} \le \frac{\sqrt{2}\beta_T \|B\|}{\sqrt{\lambda_- \tau}}.$$

Under the condition $\tau \ge \frac{8\|B\|^2\beta_T^2}{\lambda_-\Delta^2}$, the right-hand side is at most $\Delta/2$, implying $\alpha_t = 1$. A union bound over the two events completes the proof. □

D.2 Accumulation of safety-induced regret in the zero-gap case

When $\Delta = 0$, the optimal safe arm may lie on the boundary of the safety constraint, and the event $x_{t,i^*} \in X_t^s$ cannot be guaranteed uniformly. Nevertheless, the cumulative safety-induced regret can still be controlled.

Lemma D.2 (Term II bound when $\Delta = 0$)**.** *Suppose Assumptions 2.1, 2.2, and 2.4 hold. Fix any $\delta \in (0, 0.5)$ and initialize Safe-COLSTIM with $\tau \geq t_\delta$ specified in Lemma B.1. Then, with probability at least $1 - \delta$,*

$$\sum_{t=\tau+1}^{T} (1 - \alpha_t) \leq \frac{2\sqrt{2}\|B\|\beta_T(T-\tau)}{c\sqrt{\lambda_- \tau}}. \tag{38}$$

Consequently, with probability at least $1 - 2\delta$,

$$\sum_{t=\tau+1}^{T} \left(\langle x_{t,i^*}, \theta^* \rangle - \langle x_{t,i_S^*}, \theta^* \rangle\right) \leq \frac{2\sqrt{2}\|B\|\beta_T(T-\tau)}{c\sqrt{\lambda_- \tau}}. \tag{39}$$

Proof. Recall that for any $\tau < t \leq T$, with probability at least $1 - \delta$, Term II $= 1 - \alpha_t$. Thus, it suffices to establish (38).

From the definition of α_t and the fact that $\theta^\top B x_{t,i^*} \leq c$, we have

$$\alpha_t \geq \frac{c}{c + 2\beta_t \|B x_{t,i^*}\|_{M_t^{-1}}} \quad \Rightarrow \quad 1 - \alpha_t \leq \frac{2\beta_t}{c} \|B x_{t,i^*}\|_{M_t^{-1}}.$$

Applying the Rayleigh quotient bound as before yields

$$1 - \alpha_t \leq \frac{2\beta_T \|B\|}{c\sqrt{\lambda_{\min}(M_{\tau+1})}}.$$

Combining this with Lemma B.1 and summing over $t = \tau + 1, \ldots, T$ completes the proof. □

E Regret Upper Bounds for Safe-COLSTIM

In this section, we provide the detailed derivation of the problem-dependent regret bounds for Safe-COLSTIM. The main results are summarized in Theorem 4.2 and Corollary 4.3, which are stated in the main body of the paper.

Before proving Theorem 4.2 and Corollary 4.3, we specify how the probability parameters p_1, p_2, and $p_{3,t}$ are selected so that the above bounds hold. Intuitively, these parameters control the failure probabilities of the key high-probability events used throughout the analysis, namely: (i) the accuracy of the maximum likelihood estimator, (ii) the success of the initialization phase, and (iii) the concentration of the randomized perturbation mechanism.

Specifically, Lemma E.1 provides a high-probability bound for the event A_{MLE} and determines a suitable choice of p_1. Lemma E.2 characterizes the initialization event A_{init} and yields the corresponding bound p_2. Finally, a sequence of concentration arguments in the subsequent lemmas controls the event $A_{\mathrm{conc},t}$ for each round $t > \tau$, leading to the choice of $p_{3,t}$.

E.1 MLE Accuracy Bound

Lemma E.1 (Probability bound for A_{MLE})**.** *Let*

$$c_1 = \frac{2\sigma}{\mu^2}\sqrt{\frac{d}{2}\log\left(1 + \frac{8T}{d}\right) + \log\left(\frac{1}{\delta}\right)},$$

and assume that the exploration length τ satisfies $\lambda_{\min}(M_\tau) \geq 1$. Then, for any $t > \tau$, conditioning on $\|\hat{\theta}_t - \theta^*\| \leq 1$, it holds that

$$\mathbb{P}(A_{\mathrm{MLE}}) \;\geq\; 1 - p_1, \qquad \textit{where } p_1 = \frac{1}{T}.$$

Proof. For a fixed round $t > \tau$, define the vector-valued function

$$G(\theta) := \sum_{s=1}^{t-1} \big(F(\langle Z_s, \theta\rangle) - F(\langle Z_s, \theta^*\rangle)\big) Z_s.$$

By the mean value theorem, for each $s \le t-1$, there exists ξ_s lying between $\langle Z_s, \theta\rangle$ and $\langle Z_s, \theta^*\rangle$ such that

$$F(\langle Z_s, \theta\rangle) - F(\langle Z_s, \theta^*\rangle) = F'(\xi_s)\, Z_s^\top(\theta - \theta^*).$$

Recalling the definition

$$\mu := \inf\Big\{(F^*)'((x-y)^\top\theta) \;\Big|\; \|x-y\| \le 2,\ \|\theta-\theta^*\| \le 1\Big\},$$

we have $F'(\xi_s) \ge \mu$ under the conditioning $\|\hat\theta_t - \theta^*\| \le 1$.

Therefore,

$$V_t = G(\hat\theta_t) = \sum_{s=1}^{t-1} F'(\xi_s)\, Z_s Z_s^\top(\hat\theta_t - \theta^*) \ge \mu \sum_{s=1}^{t-1} Z_s Z_s^\top(\hat\theta_t - \theta^*) = \mu M_t(\hat\theta_t - \theta^*),$$

where $M_t = \sum_{s=1}^{t-1} Z_s Z_s^\top$. Consequently,

$$\hat\theta_t - \theta^* \le \mu^{-1} M_t^{-1} V_t.$$

Next, for any arm pair (i, j) at round t, by the Cauchy–Schwarz inequality,

$$\begin{aligned}
&\big|\hat u_{t,i}(X_t) - u_i^*(X_t) + u_j^*(X_t) - \hat u_{t,j}(X_t)\big| \\
&\quad = \big|(\hat\theta_t - \theta^*)^\top z_{t,i,j}\big| \le \|\hat\theta_t - \theta^*\|_{M_t}\, \|z_{t,i,j}\|_{M_t^{-1}} \\
&\quad = \sqrt{(\hat\theta_t - \theta^*)^\top M_t(\hat\theta_t - \theta^*)}\, \|z_{t,i,j}\|_{M_t^{-1}} \le \mu^{-2}\|V_t\|_{M_t^{-1}}\, \|z_{t,i,j}\|_{M_t^{-1}}. \qquad (40)
\end{aligned}$$

By Lemma A.2, with probability at least $1-\delta$, uniformly for all $t > \tau$,

$$\|V_t\|_{M_t^{-1}}^2 \le 4\sigma^2 \left(\frac{d}{2}\log\left(1 + \frac{8t}{d}\right) + \log\left(\frac{1}{\delta}\right)\right). \qquad (41)$$

Combining (40) and (41), we obtain

$$\big|\hat u_{t,i}(X_t) - u_i^*(X_t) + u_j^*(X_t) - \hat u_{t,j}(X_t)\big| \le c_1\, \|z_{t,i,j}\|_{M_t^{-1}},$$

where

$$c_1 = \frac{2\sigma}{\mu^2}\sqrt{\frac{d}{2}\log\left(1 + \frac{8T}{d}\right) + \log\left(\frac{1}{\delta}\right)}.$$

Finally, choosing $\delta = 1/T$ yields

$$\mathbb{P}(A_{\mathrm{MLE}}) \ge 1 - p_1, \qquad p_1 = \frac{1}{T}.$$

This completes the proof. □

E.2 Initialization Phase Analysis

Lemma E.2 (Probability bound for A_{init})**.** *Assume that the exploration length τ is chosen such that for all $t > \tau$,*

$$\lambda_{\min}(M_t) \;\ge\; \max\left(1,\ \frac{4\sigma^2\left(\frac{d}{2}\log\big(1 + \frac{8t}{d}\big) + \log\big(\frac{1}{\delta}\big)\right)}{\mu^2}\right).$$

Then, for any $t > \tau$, it holds that

$$\mathbb{P}(A_{\mathrm{init}}) \;\ge\; 1 - p_2, \qquad \textit{where } p_2 = \frac{1}{T}.$$

Proof. Fix an arbitrary round $t > \tau$. Define the mapping $G : \mathbb{R}^d \to \mathbb{R}^d$ as in Lemma E.1. By construction, $G(\theta^*) = 0$. Moreover, recalling the definition of the score vector,

$$V_t = \sum_{s=1}^{t-1} \epsilon_s Z_s,$$

we have

$$V_t = G(\hat{\theta}_t),$$

so that $\hat{\theta}_t$ is a solution to the equation $G(\theta) = V_t$.

Step 1: Injectivity of G. For any $\theta_1, \theta_2 \in \mathbb{R}^d$, by the mean value theorem, there exists $\bar{\theta} = v\theta_1 + (1-v)\theta_2$ for some $v \in (0,1)$ such that

$$F(\langle Z_s, \theta_1 \rangle) - F(\langle Z_s, \theta_2 \rangle) = F'(\bar{\theta})\, Z_s^\top (\theta_1 - \theta_2).$$

Consequently,

$$G(\theta_1) - G(\theta_2) = \sum_{s=1}^{t-1} F'(\bar{\theta})\, Z_s Z_s^\top (\theta_1 - \theta_2) =: E(\bar{\theta})(\theta_1 - \theta_2),$$

where

$$E(\bar{\theta}) = \sum_{s=1}^{t-1} F'(\bar{\theta})\, Z_s Z_s^\top.$$

Let μ be as defined in Assumption 2.3. Since $F' > 0$, for any $\theta_1 \neq \theta_2$,

$$\begin{aligned}(\theta_1 - \theta_2)^\top \big(G(\theta_1) - G(\theta_2)\big) &= (\theta_1 - \theta_2)^\top E(\bar{\theta})(\theta_1 - \theta_2) \\ &\geq \mu (\theta_1 - \theta_2)^\top M_t (\theta_1 - \theta_2) > 0.\end{aligned}$$

Hence, G is injective and therefore admits a well-defined inverse G^{-1}.

Step 2: Local strong monotonicity of G. For $\eta > 0$, define the Euclidean ball

$$\mathcal{B}_\eta := \{\theta : \|\theta - \theta^*\| \leq \eta\},$$

and the local slope constant

$$\mu_\eta := \inf\Big\{ (F^*)'((x-y)^\top \theta) \;\Big|\; \|x - y\| \leq 2,\ \|\theta - \theta^*\| \leq \eta \Big\}.$$

Since $\mathcal{B}_\eta$ is convex, if $\theta_1, \theta_2 \in \mathcal{B}_\eta$, then $\bar{\theta} \in \mathcal{B}_\eta$.

For any $\theta \in \mathcal{B}_\eta$, we have

$$\begin{aligned}\|G(\theta)\|^2_{M_t^{-1}} = \|G(\theta) - G(\theta^*)\|^2_{M_t^{-1}} &= (\theta - \theta^*)^\top E(\bar{\theta}) M_t^{-1} E(\bar{\theta})(\theta - \theta^*) \\ &\geq \mu_\eta^2 (\theta - \theta^*)^\top M_t (\theta - \theta^*) \geq \mu_\eta^2 \lambda_{\min}(M_t) \|\theta - \theta^*\|^2, \end{aligned} \tag{42}$$

where the last inequality follows from the Rayleigh quotient. Therefore,

$$\|G(\theta)\|_{M_t^{-1}} \geq \mu_\eta \sqrt{\lambda_{\min}(M_t)}\, \|\theta - \theta^*\|. \tag{43}$$

Step 3: Inverse image of an ellipsoid. By Lemma A of [11],

$$\big\{\theta : \|G(\theta)\|_{M_t^{-1}} \leq \mu_\eta \eta \sqrt{\lambda_{\min}(M_t)}\big\} \subseteq \mathcal{B}_\eta. \tag{44}$$

Combining (43) and (44) shows that $\|\theta - \theta^*\| \leq \eta$ implies $\theta \in \mathcal{B}_\eta$.

Step 4: Controlling the MLE error. Since $V_t = G(\hat{\theta}_t)$, inequality (43) yields

$$\|\hat{\theta}_t - \theta^*\| \le \frac{\|V_t\|_{M_t^{-1}}}{\mu_\eta \sqrt{\lambda_{\min}(M_t)}}.$$

By Lemma A.2, with probability at least $1-\delta$,

$$\|V_t\|^2_{M_t^{-1}} \le 4\sigma^2 \left(\frac{d}{2} \log\!\left(1 + \frac{8t}{d}\right) + \log\frac{1}{\delta} \right).$$

Thus,

$$\|\hat{\theta}_t - \theta^*\| \le \frac{2\sigma \sqrt{\frac{d}{2}\log\!\left(1+\frac{8T}{d}\right) + \log\frac{1}{\delta}}}{\mu_\eta \sqrt{\lambda_{\min}(M_t)}}.$$

If $\eta \le 1$, then $\mathcal{B}_\eta \subseteq \mathcal{B}_1$ and hence $\mu_\eta \ge \mu$. Under the stated condition on $\lambda_{\min}(M_t)$, the right-hand side is at most 1, which implies $\|\hat{\theta}_t - \theta^*\| \le 1$.

Finally, setting $\delta = 1/T$ yields

$$\mathbb{P}(A_{\text{init}}) \ge 1 - p_2, \qquad p_2 = \frac{1}{T}.$$

This completes the proof. □

E.3 Proof of Theorem 4.2: Problem-Dependent Bound

Proof of Theorem 4.2. We begin by describing the initialization phase. The choice of the exploration length τ must guarantee that the Gram matrix $M_{\tau+1}$ accumulated during the pure exploration phase satisfies the spectral preconditions required by both Lemma E.2 (for the initialization event A_{init}) and the subsequent MLE concentration analysis. To this end, we define the threshold round t' as

$$t' \;:=\; \max\left\{ t_\delta,\; \frac{8\sigma^2 \left(\frac{d}{2}\log\!\left(1+\frac{8T}{d}\right) + \log\frac{1}{\delta}\right)}{\mu^2 \lambda_-},\; \frac{2}{\lambda_-} \right\},$$

where t_δ and λ_- are defined in Lemma B.1 via Assumptions 2.1 and 2.4.

Choice of the initialization length. The three terms in the maximum correspond to three distinct requirements that must be simultaneously satisfied for the regret analysis to proceed:

(i) *High-probability concentration of $M_{\tau+1}$.* The first term, $t_\delta = \frac{32}{\lambda_-}\log(d/\delta)$, is the minimum exploration length required by Lemma B.1 to ensure that $\lambda_{\min}(M_{\tau+1}) \ge \frac{\lambda_- \tau}{2}$ holds with probability at least $1-\delta$. Without this term, the Matrix Chernoff bound underlying Lemma B.1 does not yield a useful guarantee.

(ii) *Sufficient signal strength for MLE concentration.* The second term, $\frac{8\sigma^2\left(\frac{d}{2}\log(1+\frac{8T}{d})+\log\frac{1}{\delta}\right)}{\mu^2\lambda_-}$, arises from the spectral precondition of Lemma E.2, which requires

$$\lambda_{\min}(M_t) \;\ge\; \frac{4\sigma^2 \left(\frac{d}{2}\log\!\left(1+\frac{8t}{d}\right) + \log\frac{1}{\delta}\right)}{\mu^2}.$$

Combined with the conclusion $\lambda_{\min}(M_{\tau+1}) \ge \frac{\lambda_-\tau}{2}$ from Lemma B.1, this requirement is met when

$$\frac{\lambda_- \tau}{2} \;\ge\; \frac{4\sigma^2 \left(\frac{d}{2}\log\!\left(1+\frac{8T}{d}\right) + \log\frac{1}{\delta}\right)}{\mu^2},$$

which, after solving for τ, gives precisely the second term in t' (using T in place of t since t' is fixed in advance and must accommodate all rounds $t \le T$).

(iii) *Invertibility of the information matrix.* The third term, $\frac{2}{\lambda_-}$, ensures that $\lambda_{\min}(M_{\tau+1}) \ge \frac{\lambda_-\tau}{2} \ge 1$, i.e., $M_t \succeq I_d$ for all $t > \tau$. This is the second precondition of Lemma E.2, required to make the confidence region C_t well-defined and to support the self-normalized concentration arguments of Appendix A.

By taking the maximum of these three terms, t' simultaneously satisfies all three requirements with a single, easily-computable threshold.

For any $\tau \geq t'$, Lemma B.1 implies that, with probability at least $1-\delta$,

$$\lambda_{\min}(M_{\tau+1}) \geq \frac{\lambda_- \tau}{2}.$$

By the definition of t', the right-hand side simultaneously satisfies

$$\frac{\lambda_- \tau}{2} \geq \max\left\{1,\ \frac{4\sigma^2\left(\frac{d}{2}\log\left(1+\frac{8T}{d}\right)+\log\frac{1}{\delta}\right)}{\mu^2}\right\}.$$

Hence the spectral conditions required by Lemma E.2 hold.

The spectral lower bound on $M_{\tau+1}$ above provides the precondition required by Lemma E.2 for the initialization event A_{init}. Moreover, the Hessian matrix

$$H_t \ := \ \sum_{s=1}^{t-1} (F^*)'(Z_s^\top \theta)\, Z_s Z_s^\top$$

satisfies $H_t \succeq \mu M_t \succ 0$ for all $t > \tau$ by Assumption 2.3 (which guarantees $(F^*)'(Z_s^\top\theta) \geq \mu$ in the relevant parameter neighborhood). This ensures that H_t is positive definite throughout the safe exploration–exploitation phase, supporting the local strong monotonicity argument in the proof of Lemma E.2.

Combining Lemmas E.1 and E.2, we obtain

$$\mathbb{P}(A_{\text{init}}) \geq 1-p_2, \qquad \mathbb{P}(A_{\text{MLE}}) \geq 1-p_1,$$

with $p_1 = p_2 = 1/T$.

Concentration event probability. We now derive a suitable choice of $p_{3,t}$. Fix an arbitrary round $t > \tau$ and condition on the history $\mathcal{H}_{t-1}$. For notational simplicity, we suppress this conditioning when it is clear. Recall (29), we decompose the probability of its complement using the Bernoulli coupling variable B_t with $\mathbb{P}(B_t = 1) = p_t$:

$$\mathbb{P}\big(A^c_{\text{conc},t}\big) = \mathbb{P}\big(A^c_{\text{conc},t} \mid B_t = 1\big)\,\mathbb{P}(B_t = 1) + \mathbb{P}\big(A^c_{\text{conc},t} \mid B_t = 0\big)\,\mathbb{P}(B_t = 0).$$

The event inside the probability is trivially bounded by 1 when $B_t = 1$, and thus

$$\mathbb{P}\big(A^c_{\text{conc},t}\big) \leq p_t + \mathbb{P}(\exists\{i,j\} \subset A : \\ \left|\tilde{u}_{t,i}(X_t) - \tilde{u}_{t,j}(X_t) - \hat{u}_{t,i}(X_t) + \hat{u}_{t,j}(X_t)\right| > c_2\|z_{t,i,j}\|_{M_t^{-1}} \ \Big|\ B_t = 0\Big).$$

Under the event $B_t = 0$, all perturbations are identical: for every $i \in A$,

$$\epsilon_{t,i} = \epsilon := \min\big(C_{\text{thresh}},\, \max(-C_{\text{thresh}}, \tilde{\epsilon})\big), \qquad \tilde{\epsilon} \sim G.$$

Therefore,

$$\mathbb{P}\big(A^c_{\text{conc},t}\big) \leq p_t + \mathbb{P}\Big(|\epsilon|\,\big|\|x_{t,i}\|_{M_t^{-1}} - \|x_{t,j}\|_{M_t^{-1}}\big| > c_2\|z_{t,i,j}\|_{M_t^{-1}} \ \Big|\ B_t = 0\Big).$$

Using the reverse triangle inequality,

$$\big|\|x_{t,i}\|_{M_t^{-1}} - \|x_{t,j}\|_{M_t^{-1}}\big| \leq \|x_{t,i} - x_{t,j}\|_{M_t^{-1}} = \|z_{t,i,j}\|_{M_t^{-1}},$$

we further obtain

$$\mathbb{P}\big(A^c_{\text{conc},t}\big) \leq p_t + \mathbb{P}(|\epsilon| > c_2 \mid B_t = 0).$$

Since $|\epsilon| \leq C_{\text{thresh}} < c_2$, the second term vanishes, and hence

$$\mathbb{P}\big(A^c_{\text{conc},t}\big) \leq p_t.$$

Consequently, we set

$$p_{3,t} = p_t = \min\left(1,\ \frac{\sqrt{2d}}{2\sqrt{t-\tau}}(3c_1 + c_2)\sqrt{\log\left(\frac{2T}{d}\right)}\right),$$

which ensures

$$\mathbb{P}(A_{\text{conc},t} \mid \mathcal{H}_{t-1}) \geq 1 - p_{3,t}.$$

Combining regret terms. Summing over t, we obtain

$$\sum_{t=\tau+1}^{T} p_{3,t} \le \frac{\sqrt{2d}}{2}(3c_1+c_2)\sqrt{\log\Big(\frac{2T}{d}\Big)}\sum_{t=1}^{T}\frac{1}{\sqrt{t}} \le \frac{\sqrt{2d}}{2}(3c_1+c_2)\sqrt{T\log\Big(\frac{2T}{d}\Big)} = \mathcal{O}\Big(d\sqrt{T}\log T\Big).$$

Applying Theorem C.1, for any $c_3 \ge \sqrt{2d}$,

$$\mathbb{E}[R_T^a(\text{Term I})] \le \Delta_{\max}\tau + \Delta_{\max}(p_1+p_2)T + \Delta_{\max}\sum_{t=\tau+1}^{T} p_{3,t} + \frac{c_3}{2}(3c_1+c_2)\sqrt{T\log\Big(\frac{2T}{d}\Big)}. \tag{45}$$

Moreover, by Lemma D.1,

$$\sum_{t=\tau+1}^{T}\big(\langle x_{t,i^*},\theta^*\rangle - \langle x_{t,i_S^*},\theta^*\rangle\big) \le 0,$$

and trivially

$$\sum_{t=1}^{\tau}\big(\langle x_{t,i^*},\theta^*\rangle - \langle x_{t,i_S^*},\theta^*\rangle\big) \le \Delta_{\max}\tau.$$

Thus,

$$\mathbb{E}[R_T^a(\text{Term II})] \le \Delta_{\max}\tau. \tag{46}$$

Combining (45) and (46) yields

$$\mathbb{E}[R_T^a(\text{Safe-COLSTIM})] \le 2\Delta_{\max}\tau + \Delta_{\max}(p_1+p_2)T + \Delta_{\max}\sum_{t=\tau+1}^{T} p_{3,t} + \frac{c_3}{2}(3c_1+c_2)\sqrt{T\log\Big(\frac{2T}{d}\Big)}.$$

Finally, choosing

$$\tau \ge T_\Delta \vee t', \qquad t' = \mathcal{O}(d\log T),$$

and using $\beta_T = \mathcal{O}(\sqrt{d\log T})$ and $c_2 \le c_1$, we conclude that

$$\mathbb{E}[R_T^a(\text{Safe-COLSTIM})] = \mathcal{O}\Big(d\sqrt{T}\log T\Big),$$

which completes the proof. □

E.4 Proof of Corollary 4.3: Improved Bound under Spectral Growth

Proof of Corollary 4.3. We adopt the same choices of constants and failure probabilities as in Theorem 4.2. The key difference is that we now invoke the second statement of Theorem C.1.

Specifically, suppose that

$$\sum_{t=\tau+1}^{T}\lambda_{\min}^{-1/2}(M_t) \le b\sqrt{T},$$

for some constant $b > 0$. Then, for any constant $c_3 \ge \sqrt{2}\,b$, Theorem C.1 implies

$$\mathbb{E}[R_T^a(\text{Term I})] \le \Delta_{\max}\tau + \Delta_{\max}(p_1+p_2)T + \Delta_{\max}\sum_{t=\tau+1}^{T} p_{3,t} + \frac{c_3}{2}(3c_1+c_2)\sqrt{T\log\Big(\frac{2T}{d}\Big)}.$$

Recall from Theorem 4.2 that $c_2 \le c_1 = \mathcal{O}\big(\sqrt{d\log T}\big)$ and that the initialization length satisfies $\tau = o(\sqrt{dT})$. Moreover, under the spectral assumption, we have

$$\sum_{t=\tau+1}^{T} p_{3,t} \le \frac{\sqrt{2}\,b}{2}(3c_1+c_2)\sqrt{T\log\Big(\frac{2T}{d}\Big)} = \mathcal{O}\Big(\sqrt{dT}\log T\Big).$$

In addition, $p_1 = p_2 = 1/T$ as established in Lemmas E.1 and E.2.

Combining the above bound on Term I with the analysis of Term II from (46), we obtain

$$\mathbb{E}[R_T^a(\text{Safe-COLSTIM})] = \mathcal{O}\Big(\sqrt{dT}\log T\Big),$$

which completes the proof. □

E.5 Proof of Theorem 4.4: Worst-Case Bound

Proof of Theorem 4.4. Invoking Theorem C.1, for any constant $c_3 \geq \sqrt{2d}$, we obtain

$$\mathbb{E}[R_T^a(\text{Term I})] \leq \Delta_{\max}\tau + \Delta_{\max}(p_1+p_2)T + \Delta_{\max}\sum_{t=\tau+1}^{T} p_{3,t} + \frac{c_3}{2}(3c_1+c_2)\sqrt{T\log\left(\frac{2T}{d}\right)}.$$

On the other hand, by Lemma D.2, we have

$$\sum_{t=\tau+1}^{T}\Big(\langle x_{t,i^*},\theta^*\rangle - \langle x_{t,i_S^*},\theta^*\rangle\Big) \leq \frac{2\sqrt{2}\,\|B\|\beta_T(T-\tau)}{c\sqrt{\lambda_-\,\tau}}.$$

Moreover, during the initialization phase,

$$\sum_{t=1}^{\tau}\Big(\langle x_{t,i^*},\theta^*\rangle - \langle x_{t,i_S^*},\theta^*\rangle\Big) \leq \Delta_{\max}\tau.$$

Combining the above two bounds yields

$$\mathbb{E}[R_T^a(\text{Term II})] \leq \Delta_{\max}\tau + \frac{2\sqrt{2}\,\|B\|\beta_T(T-\tau)}{c\sqrt{\lambda_-\,\tau}}. \tag{47}$$

Putting together the bounds on Term I and Term II, we obtain

$$\begin{aligned}\mathbb{E}[R_T^a(\text{Safe-COLSTIM})] \leq\ & 2\Delta_{\max}\tau + \Delta_{\max}(p_1+p_2)T \\ &+ \Delta_{\max}\sum_{t=\tau+1}^{T} p_{3,t} + \frac{2\sqrt{2}\,\|B\|\beta_T(T-\tau)}{c\sqrt{\lambda_-\,\tau}} + \frac{c_3}{2}(3c_1+c_2)\sqrt{T\log\left(\frac{2T}{d}\right)}.\end{aligned}$$

Recall that

$$\lambda_- = \frac{2\epsilon^2}{d\|B\|^2}, \qquad \beta_T = \mathcal{O}\Big(\sqrt{d\log T}\Big).$$

Choosing

$$\tau = \mathcal{O}\Big((dT)^{2/3}\sqrt{\log T}\Big),$$

we obtain

$$\frac{2\sqrt{2}\,\|B\|\beta_T(T-\tau)}{c\sqrt{\lambda_-\,\tau}} = \mathcal{O}\Big((dT)^{2/3}\sqrt{\log T}\Big).$$

Furthermore, using the same analysis as in Theorem 4.2,

$$\sum_{t=\tau+1}^{T} p_{3,t} = \mathcal{O}\Big(d\sqrt{T}\log T\Big).$$

By arguments analogous to those in the proof of Theorem 4.2, we conclude that the expected cumulative average regret of Safe-COLSTIM satisfies

$$\mathbb{E}[R_T^a(\text{Safe-COLSTIM})] = \mathcal{O}\Big(d\,T^{2/3}\log T\Big).$$

□

E.6 Proof of Corollary 4.5: Improved Worst-Case Bound

Proof of Corollary 4.5. We adopt the same setup as in Theorem 4.4. The key difference is that we now invoke the second statement of Theorem C.1.

Specifically, suppose that

$$\sum_{t=\tau+1}^{T}\lambda_{\min}^{-1/2}(M_t) \leq b\sqrt{T},$$

for some constant $b > 0$. Then, for any constant $c_3 \geq \sqrt{2}\, b$, Theorem 4.4 implies

$$\mathbb{E}[R_T^a(\text{Safe-COLSTIM})] \leq \; 2\Delta_{\max}\tau + \Delta_{\max}(p_1 + p_2)T$$
$$+ \Delta_{\max} \sum_{t=\tau+1}^{T} p_{3,t} + \frac{2\sqrt{2}\,\|B\|\beta_T(T-\tau)}{c\sqrt{\lambda_-\,\tau}} + \frac{c_3}{2}(3c_1 + c_2)\sqrt{T\log\left(\frac{2T}{d}\right)}.$$

Recall from Theorem 4.4 that $c_2 \leq c_1 = \mathcal{O}\big(\sqrt{d\log T}\big)$ and that

$$\tau = \mathcal{O}\Big((dT)^{2/3}\sqrt{\log T}\Big)\,.$$

Moreover, under the assumed spectral growth condition,

$$\sum_{t=\tau+1}^{T} p_{3,t} \leq \frac{\sqrt{2}\,b}{2}(3c_1 + c_2)\sqrt{T\log\left(\frac{2T}{d}\right)} = \mathcal{O}\Big(\sqrt{dT}\log T\Big)\,,$$

and $p_1 = p_2 = 1/T$ as established in Lemmas E.1 and E.2.

Combining the above bounds with the analysis of $\mathrm{Term\ II}$ from (47), we conclude that

$$\mathbb{E}[R_T^a(\text{Safe-COLSTIM})] = \mathcal{O}\Big((dT)^{2/3}\log T\Big)\,,$$

which completes the proof. □

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